

Williams Cycle effects and successive cyclicity unified

Abstract

The Williams Cycle (Williams 2003) generalizes the correlation between a clause's relative size and its selective opacity. Most previous accounts (Keine 2020, Poole 2022; Meadows 2023) maintain Phase Theory (Chomsky 2001) to explain reflexes of successive cyclicity, and propose an independent mechanism to derive Williams Cycle effects. I propose a unified analysis of Williams Cycle effects and successive cyclicity. Under my unified constraint, probing must proceed along well-formed functional-sequence fragments. When cross-clausal sub-extraction proceeds successive cyclically, Williams Cycle effects are obviated. When successive cyclicity is unavailable, the clause's movement to an equal-sized matrix projection feeds sub-extraction, but sub-extraction obeys the Williams Cycle.

Keywords: Williams Cycle, size-dependent opacity, successive cyclicity, hyperraising

1 Introduction

The Williams Cycle (Williams 1974, 2003, 2013) captures the attested patterns of *selective opacity* (term from Keine 2016, 2019, 2020). The best known illustration of the Williams Cycle is the ban on *hyperraising* from English CPs. First of all, English CPs are not an islands: *wh*-movement can proceed from them and land in matrix Spec,CP, as in (1).

- (1) [CP [Which country]₁ did you think [CP ₁ was defeated in the Battle of Mohács]]?

However, the English CP is selectively opaque to certain movement types: Raising to Spec,TP cannot proceed from CPs, as illustrated in (2). That is, English CPs are opaque to the syntactic phenomenon that has been referred to as *hyperraising* (Chomsky 1973).

- (2) *[TP [The Kingdom of Hungary]₁ seemed [CP ₁ was defeated in the Battle of Mohács]].

In order to have a grammatical raising structure in English, the embedded clause needs to maximally be a TP, as in (3).

- ↓-----↑
- (3) [_{TP} [The Kingdom of Hungary]₁] seemed [_{TP} ₁ to be defeated in the Battle of Mohács]].

The opacity of CPs to movement landing within TP (Spec,TP) falls under the Williams Cycle in (4) assuming C>T in the *f*(unctional) *seq*(uence). In this paper, I am treating the Williams Cycle as a descriptive generalization.

- (4) **Williams Cycle:** Within the current γ P, a syntactic operation may not target an element across β P, where β is higher than γ in the functional sequence.
(based on Williams 2003; Poole 2022:p.383)

Besides English, Williams Cycle effects have been attested amongst others in German (Müller 2014a,b), Hindi (Keine 2016, 2019, 2020), Swahili (Meadows 2023), Georgian (Bondarenko 2025), Mandarin Chinese (Meadows and Yan 2024; but see Chen and Yip 2026), Hungarian (Egressy 2023, 2025, 2026), and Mongolian (Lim 2025, 2026). At the same time, there are some well-known examples in which Williams Cycle effects are *obviated*. For instance, while movement from CPs cannot land in Spec,TP in English, this movement step is grammatical in Buryat (Bondarenko 2017b), Brazilian Portuguese (Nunes 2008), and Zulu (Halpert 2019).

The tension that motivates the discussion in this paper is that Standard Phase Theory (Chomsky 2000, 2001), which is currently the standard explanation behind the attested reflexes of *successive cyclic* movement (Chomsky 1973; 1977; 1981; McCloskey 1979; a.o.), cannot derive a well-attested phenomenon, Williams Cycle effects (Müller 2014a,b; Keine 2016, 2019, 2020; Poole 2022; Meadows 2023; Cao 2025).¹ Under Phase Theory, certain constituents are *phases* (e.g. CPs are phases), which has consequences for their opacity: The complement of a phase head is uniformly inaccessible for external probes, whereas the edge of a phase is uniformly accessible for external probes. Therefore, if DP moves in (5) to Spec,CP, Phase Theory itself predicts that it can move to *any* matrix position including Spec,TP, which is a violation of the Williams Cycle in (4) and ungrammatical in languages like English, as exemplified in (2). In general, since successive cyclicity makes

clauses uniformly transparent, Phase Theory cannot itself cannot derive selective opacity and Williams Cycle effects.

$$(5) \quad [_{TP} \checkmark \dots [_{CP} DP [\text{C} [\dots DP \dots]]]]$$

Previous accounts introduce *secondary constraining mechanisms* to derive Williams Cycle effects, but they keep using Phase Theory (or some earlier related mechanism) for the derivation of successive cyclicity (e.g. Chomsky 1973; Müller 2014a,b; Keine 2016, 2019, 2020; Poole 2022; Meadows 2023). Consequently, a certain constituent under these accounts can be opaque for a certain movement type for *two independent reasons*: Under one of the constraining mechanisms, it can be phase that cannot be escaped from successive cyclically; or, under the other constraining mechanism, it can be too large or let through only certain movement types (i.e. it may exhibit Williams Cycle effects).

In this paper, I propose that we do not need separate constraining mechanisms on syntactic locality; we can overcome this patchwork of locality constraints. Williams Cycle effects, their obviation, and successive cyclicity can be derived from a *unified constraint*. In particular, I propose the constraint *Probe Along Fseq-Fragments* (PAFF) (following the intuition of Egressy 2023), which requires probing to proceed along a sequence of categories that constitute a well-formed fragment of some functional sequence. Under the PAFF-based theory, sub-extraction from a clause can happen via two strategies: (i) via successive cyclic movement through edge of the clause, as in (6); (ii) or after the movement of the entire clause to an equal-sized matrix projection, as in (7).

$$(6) [_{CP} \checkmark [_{TP} \checkmark \dots [_{CP} XP \dots XP \dots]]] \quad (7) [_{CP} \checkmark [_{TP} \times \dots \{_{CP} \dots XP \dots \} \dots] \{_{CP} \dots YP \dots \}_{CP}]$$

Successive cyclic movement will be argued to obviate Williams Cycle effects, for example, movement from CP can land in Spec,TP in (6). In contrast, Williams Cycle effects will be argued to arise in cases where a certain clause cannot be escaped successive cyclically but only after clausal movement to an equal-sized matrix projection, for example,

movement from CP cannot land in Spec,TP in (7). That is, the PAFF-based theory predicts that Williams Cycle effects and successive cyclicity never co-occur. A further general prediction is that if opacity is selective, it follows the Williams Cycle pattern, whereas the obviation of Williams Cycle effects always results in *no opacity*.

In Section 2, I motivate PAFF, the new constraint, which unifies domain-based syntactic locality. Section 3 discusses how successive cyclic movement is an escape strategy under PAFF and how Williams Cycle obviation arises with successive cyclicity. Section 4 presents clausal movement as the alternative strategy for PAFF-compliant sub-extraction and shows how Williams Cycle effects arise under this strategy. Section 5 discusses the third option, when neither successive cyclicity nor clausal movement is available so the embedded clause remains opaque. Finally, Section 6 discusses how the predictions of the PAFF-based theory differ from those of *phase unlocking* (Rackowski and Richards 2005; Den Dikken 2009; Halpert 2012; Van Urk and Richards 2015), which is another theory associating the possibility of sub-extraction from a clause with its movement.

2 Probe Along *Fseq*-Fragments

2.1 Previous accounts: Phase theory and something else

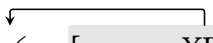
Standard Phase Theory (Chomsky 2000, 2001), which is currently the standard explanation behind successive cyclicity, cannot derive Williams Cycle effects and selective opacity in general (Müller 2014a,b; Keine 2016, 2019, 2020; Poole 2022; Meadows 2023; Cao 2025). Under Phase Theory (Chomsky 2000, 2001), the locality constraint deriving successive cyclic movement is the Phase-Impenetrability Condition (PIC), which is defined in (8).

- (8) **Phase-Impenetrability Condition:** The domain of H is not accessible to operations outside HP; only H and its edge are accessible to such operations. (Chomsky 2000:p.13).

If a certain category H is phasal (e.g. C is standardly assumed to be phasal), the complement of H is inaccessible for probes on heads c-commanding HP, as in (9).

- (9)
$$\left[\times \dots \left[\text{HP}_{\text{phase}} \checkmark \left[\text{H} \left[\dots \text{XP} \dots \right] \right] \right] \right]$$

At the same time, Spec,HP, the *edge* of some HP phase, is uniformly accessible for external probes, as shown in (10). Therefore, an XP that moves successive cyclically through Spec,HP becomes accessible by external probes.

- (10)  [✓ ... [HP_{phase} XP [H [...]]]]

Phase Theory relying only on the PIC in (8) cannot derive Williams Cycle effects (Williams Cycle repeated in (11)) because the opacity of phases is never selective: any XP that is originally phase-internal and inaccessible for external probes can move to the phase edge and become accessible (for specific previous discussions see Müller 2014a: p.123 and Section 1.2 in Keine 2016 but also see Müller 2014b; Keine 2019, 2020; Poole 2022; Egressy 2023; Meadows 2023; Cao 2025).

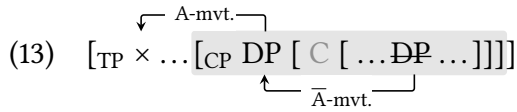
- (11) **Williams Cycle:** Within the current γ P, a syntactic operation may not target an element across β P, where β is higher than γ in the functional sequence.
(based on Williams 2003; Poole 2022:p.383)

For example, in (12), if a DP moves to Spec,CP, Phase Theory itself does not exclude its subsequent movement to Spec,TP. While this is the correct prediction for Buryat, Brazilian Portuguese, and Zulu, it is incorrect for English, as discussed in the Introduction.

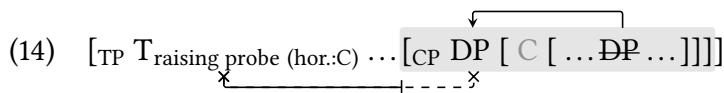
- (12)  [TP ✓ ... [CP DP [C [... ~~DP~~ ...]]]]

For this reason, previous accounts needed to enrich the theory of syntactic locality with additional constraining mechanisms to derive selective opacity and Williams Cycle effects while also keeping phases and the PIC as the explanation for successive cyclicity. Here I will briefly review three approaches. Accounts building on the Ban on Improper movement intuition (Chomsky 1973, 1981; May 1979) made use of the distinction between various *movement types*. For example, while movement to Spec,CP is an instance of $\bar{A}$ -movement in English, movement to the Spec,TP subject position is an instance A-movement. Ban on Improper Movement accounts put constraints on the order of movement types that

can affect a certain constituent. Thus, (13) is ruled out because successive cyclic $\bar{A}$ -movement to Spec,CP cannot be followed by A-movement to Spec,TP (for a more fine-grained movement-type hierarchy, see Abels 2007). Under these accounts, one way to explain the grammaticality of hyperraising from Spec,CP in a certain language is to say that the movement step to Spec,CP in that language is not $\bar{A}$ -movement but A-movement (for details, see Fong 2019).



In Keine's (2016; 2019; 2020) account, phases deriving successive cyclicity co-exist with *horizons*, a probe-constraining mechanism deriving selective opacity. Under this account, a probe may be specified for a category –its horizon– that blocks its search. For example, if the probe of subject raising in English has the category C as a horizon, the search of this probe is blocked by CPs, as shown in (14). That is, English subject raising probes cannot raise DPs from CPs, whether they have moved to Spec,CP or not. What horizons the various probes have changes on a language-by-language basis. Hence, in the horizon-based theory, hyperraising to Spec,TP is possible in a language if the raising probe on T does not have C as a horizon.



Following the spirit of Williams' original work (Williams 2003), Poole (2022), Meadows (2023), and Lim (2025, 2026) derive Williams Cycle effects by constraining *the time of embedding*: Matrix and embedded clauses are built in parallel work spaces, and embedded clauses are embedded in the matrix structure only at the point when the matrix clause reaches their size. As shown in (15), CP-sized complement clauses "occur" in the matrix VP only when the matrix C merges with the matrix TP. TP-sized clauses are embedded earlier, when the matrix T merges with the matrix ν P. Before clauses get embedded, only a placeholder occurs in the matrix structure. Therefore, these accounts implement Williams

Cycle as the size-dependent substitution of the embedded clause: In (15), when the raising probe on the matrix T tries to probe down, the embedded CP is simply absent of the structure (rather than being just opaque), because only the placeholder of the embedded CP is in the matrix VP.

$$\begin{aligned}
 (15) \quad & [\text{VP } V \text{ (CP's placeholder)}] \rightarrow \\
 & [\text{vP } v[\text{VP } V \text{ (CP's placeholder)}]] \rightarrow \\
 & [\text{TP } T_{\text{raising probe}} [\text{vP } v[\text{VP } V \text{ (CP's placeholder)}]]] \rightarrow \\
 & [\text{CP} [\text{C} [\text{TP } \text{DP } T_{\text{vP}} v[\text{VP } V [\text{CP} [\text{C} [\text{TP } T_{\text{vP}} v[\text{VP } V [\text{.....}]]]]]]]]]]
 \end{aligned}$$

While the delayed substitution mechanism derives Williams Cycle effects, it does not derive successive cyclicity, that is, successive cyclicity needs to be derived from some other mechanism (Poole 2022; Meadows 2023). Among currently mainstream theories, Phase Theory is a good candidate for this task. Space does not permit to discuss how hyperraising can be derived by a delayed substitution account; for a solution, see Meadows (2023).

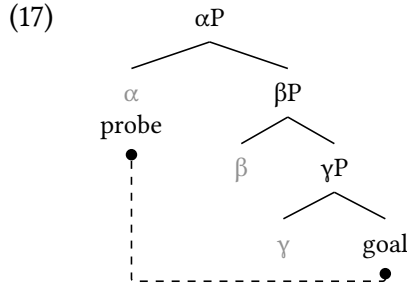
In sum, all these existing accounts make use one mechanism (e.g. Phase Theory) to derive successive cyclicity and propose a secondary constraining mechanism to derive Williams Cycle effects.

2.2 A unified constraint

In this enterprise, I propose that we do not need multiple constraining mechanisms, Phase Theory plus *something else*, to account for the various patterns of syntactic locality. Instead, I propose that the PIC and any existing secondary constraining mechanism deriving Williams Cycle effects should be subsumed by a single locality constraint, *Probe Along Fseq-Fragments* (PAFF), defined in (16).² I argue in this paper that PAFF *on its own* is capable of deriving Williams Cycle effects, Williams Cycle obviation, and successive cyclicity in a unified system, without further constraints on domain-based locality.³

- (16) **Probe Along *Fseq*-Fragments (PAFF):** Within the domain dominated by the maximal projection of the head that the probe is located on, the sequence of category labels of the heads c-commanding the goal must correspond to a well-formed fragment of some functional sequence. (based on Egressy 2023)

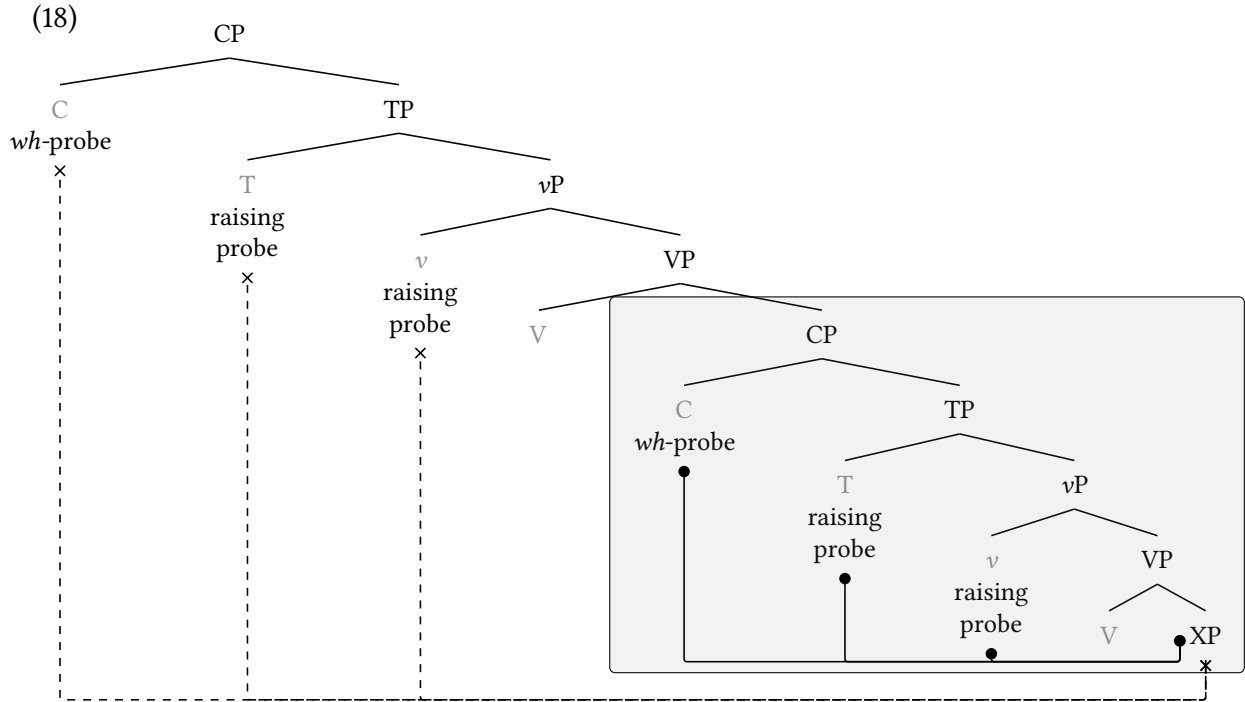
The tree diagram in (17) shows how PAFF in (16) constrains probing given that the probe is located on the head α and the goal is complement of the head γ . Since the probe is located on α , the maximal projection within which we need to consider the heads c-commanding the goal is αP . The heads c-commanding the goal within this domain from the highest to the lowest are α , β , and γ . The probing along the α - β - γ sequence is PAFF-compliant iff i) α , β , and γ belong to the same *fseq*; ii) in this *fseq*, α is higher than β and β is higher than γ (i.e. $\alpha > \beta > \gamma$); iii) α , β , γ are adjacent in this *fseq* (i.e. there is no δ in the *fseq* such that $\alpha > \delta > \beta$ or $\beta > \delta > \gamma$).



I assume that in extended projections, the presence of a higher *fseq*-category implies the presence of *all* lower *fseq*-categories. For example, if a clause has a CP layer headed by C, this clause must have a TP layer, a *v*P layer, and a VP layer, too.

PAFF formalizes an intuition which was originally formulated from Chomsky (1977:p.85): In some sense, operations must be internal to clauses or, more generally, to extended projections (clauses are extended projections; see Grimshaw 2000). Whether an operation is local in this sense is tracked by the heads of extended projections. Another way of thinking of PAFF is that the search of a probe can proceed past lower and lower projections as long as their categories are the downward continuation of the *fseq*.

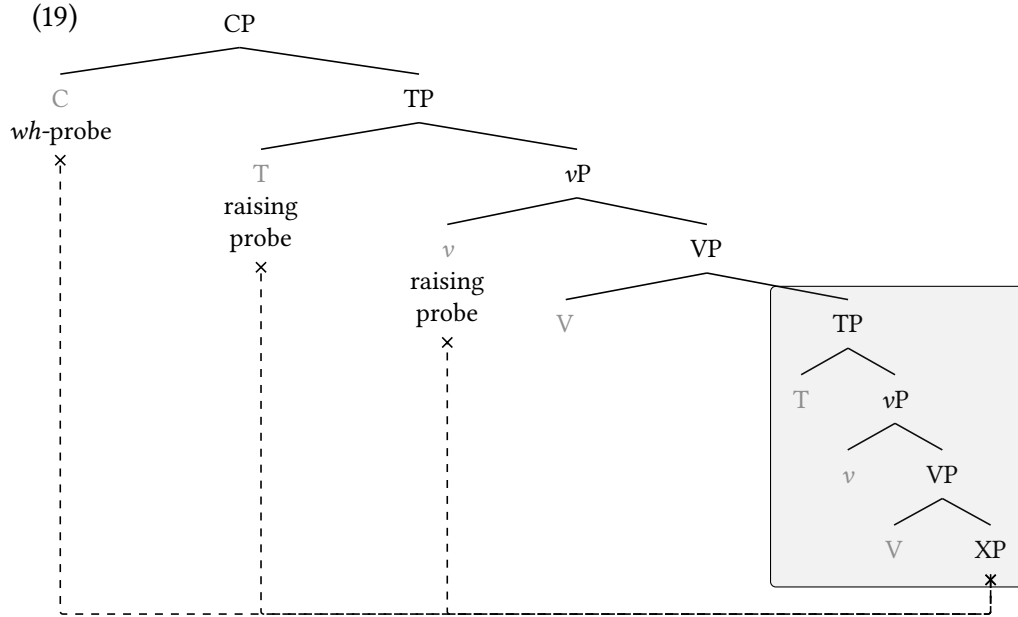
As shown by the solid lines in the schematic tree diagram in (18), probes internal to the embedded CP can uniformly reach XP in the embedded VP. Under the *fseq* C>T>*v*>V, the *wh*-probe on the embedded C probes along the well-formed C-T-*v*-V sequence; the raising probe on the embedded T probes along the well-formed T-*v*-V sequence; and the raising probe on the embedded *v* probes along the well-formed *v*-V sequence. In short, movement to Spec,CP as well to Spec,TP and Spec,*v*P can happen within the CP in (18).



In general, PAFF in (16) does not allow probes on matrix heads to probe cross-clausally and reach goals occurring inside VP-internal embedded clauses: In terms of positions in the *fseq*, the highest head of a complement clause does not serve as a downward continuation for the sequence of matrix heads. Thus, in (18), the probes on matrix heads cannot reach XP in the embedded CP, as the dashed lines show: The *wh*-probe on the matrix C cannot probe along the ill-formed C-T-*v*-V-C-T-*v*-V sequence; the raising probe on the matrix T cannot probe along the ill-formed T-*v*-V-C-T-*v*-V sequence; and the raising probe on the matrix *v* cannot probe along the ill-formed *v*-V-C-T-*v*-V sequence. (PAFF-violating subsequences

are underlined throughout the paper.) In particular, these instances of probing are excluded by PAFF because probing proceeds along the V-C sub-sequence, which is problematic because V and C are not adjacent in the *fseq* and $C > V$. In short, the movement of XPs cannot proceed from the embedded CP to matrix Spec,CP, Spec,TP, or Spec,vP.

As shown by (19), TP-sized clauses are also opaque to cross-clausal operations: The *wh*-probe on the matrix C cannot probe along the ill-formed C-T-*v*-V-T-*v*-V sequence; the raising probe on the matrix T cannot probe along the ill-formed T-*v*-V-T-*v*-V sequence; and the raising probe on the matrix *v* cannot probe along the ill-formed T-*v*-V-T-*v*-V sequence. Similar to the (18), the problem arises in these cases between the matrix V and the embedded T in each of these: V and T are not adjacent in the *fseq* and $T > V$.



In general, the lowest head V in the matrix clause is never higher in the *fseq* than the highest head of the complement clause because V is the lowest head in the *fseq*. Consequently, the maximal projections of (clausal) extended projections function as locality boundaries in the context of probing under PAFF. Hence, with vP-sized complement clauses, probing cannot cross the ...V-*v*... sequence because V is lower than *v*; for VP-sized clauses,

probing cannot cross the ...V-V... sequence because no *fseq* contains two Vs. In short, PAFF rules out operations directly crossing multiple extended projections (e.g. clauses).

Following the intuition by (Chomsky 1977:p.85), operations (apparently) crossing multiple extended projections (e.g. some XP is base generated in the embedded clause but ends up in a matrix position) emerge via various *escape strategies*. These escape strategies make cross-clausal operations PAFF-compliant by adding secondary movement steps that make goals occur in the same *fseq*-defined domain as the probe. The options for the opacity profiles arise from the availability/unavailability of these various secondary movement steps. I propose that any given extended projection in any language (i.e. any given "βP") exhibits one of the three options represented in (20).

		Successive cyclicity	Moving clauses	Opacity pattern
(20)	Option 1	✓	(irrelevant)	No opacity
	Option 2	×	✓	Obeys Williams Cycle
	Option 3	×	×	Complete opacity

Under Option 1, the extended projection can be escaped via successive cyclicity, that is, the secondary movement affects the goal itself. As shown in Section 3, upon successive cyclic movement to the highest specifier/*edge* of some extended projection, the goal can be accessed from matrix heads along exclusively matrix heads, that is, matrix probes do not need to violate PAFF by crossing embedded heads. This option will be exemplified by the English T(P) and the P'urhepecha C(P). Under Option 2, the highest specifier of the extended projection cannot be used for successive cyclic movement, so the goal itself cannot undergo the secondary movement. However, the extended projection can still be escaped if the whole extended projection (clause) undergoes secondary movement to an equal-size matrix projection. In Section 4, it will be argued for on the basis of German and Dutch CPs how this re-merger with an equal-sized matrix projection forms a well-formed *fseq*-fragment from matrix and embedded heads and hence enables PAFF-compliant sub-extraction. Finally, it is also an option (Option 3) that it is neither the case that the goal can undergo successive cyclic movement to the edge of the extended projection, nor the case

that the extended projection can undergo movement to an equal-sized matrix projection. In this case, the extended projection is completely opaque to cross-clausal subextraction. One candidate for this setup comes from Tsez, as discussed in Section 5.

It needs to be emphasized that the PAFF-based theory does not make use of the concept of phases; more generally, domainhood is not established independently of probing and is not associated with certain categories (e.g. Cs). Therefore, any given extended projection, even the ones not typically thought of as phasal (e.g. TPs or ForcePs), exhibits to one of the options in (20). Which option of the three options a given β P extended projection (clause) exhibits in a given language is a point of parametric variation, which is encoded in the lexicon of the language, more specifically, in features associated with the category β in that language. Hence, it can vary language-internally which option(s) the individual categories exhibit (e.g. English C differs from English T) and it can vary cross-linguistically which option a certain category exhibits (e.g. German C differs from P'urhepecha C).

A central claim of this paper is that if sub-extraction from a given embedded β P in a given language happens via the movement of β P to an equal-sized matrix projection (i.e. this β P exhibits Option 2), this β P in this language exhibits Williams Cycle effects. In other words, Williams Cycle effects arise in the absence of successive cyclicity and in the presence of clausal movement. In contrast, if a given embedded β P in a given language can be escaped successive cyclically (i.e. this β P exhibits Option 1), Williams Cycle effects are obviated for sub-extraction from this β P in this language. Since there is no sub-extraction under Option 3, sub-extraction from β Ps of Option 3 trivially obeys the Williams Cycle.

From a certain point of view, PAFF also implements the intuition of Bošković (2014): Extended projections and *fseqs* are the basic units of syntactic locality, opposing the more traditional view that associates phasality/domainhood with certain *categories* (e.g. C and D are phasal and phases are CP- and DP-sized). Another similar idea in the literature is the *buffer*-based system by Müller (2014a,b). Müller proposes that every phrase is a

phase, so the movement of any XP must make a successive cyclic stop in every projection until it reaches its final position. At every successive cyclic step, the category label of the projection is recorded into a *buffer*; every buffer needs to correspond to a well-formed *fseq*-fragment in the end of the derivation. For example, a *wh*-object moving to Spec,CP ends up with the buffer V-*v*-T-C, which is a well-formed buffer. At the same time, it needs to be highlighted these previous accounts do not abandon phases from the theory: Bošković (2014) defines the highest projections of extended projections as *phasal* boundaries and Müller (2014a,b) derives successive cyclicity from the phasality of projections. Additionally, Müller (2014a,b) has the same drawback as the accounts discussed in the previous section: Besides phases, it has a secondary constraining mechanism, buffers, to derive selective opacity. In contrast, the PAFF-based theory does not need phases and the PIC.

3 Successive cyclicity obviates Williams Cycle effects

3.1 Criterial features, edge features, and their distribution

Building on the intuitions of Rizzi (2004), Chomsky (2013), Georgi (2014, 2017), I distinguish between two kinds of movement features. Any movement feature is either a *criterial feature* denoted as $[\odot F \odot]$ and defined in (21) or an *edge feature* denoted as $[\odot EF \odot]$ and defined in (22). Features driving external merge (e.g. between the categories of an extended projection) are denoted as $[\bullet F \bullet]$, features driving Agree are denoted as $[\star F \star]$ (Heck and Müller 2007).

(21) **Criterial features:**

1. $[\odot F \odot]$ on β attracts an XP bearing $[F]$.
2. Subsequently, XP cannot move out Spec, β P, i.e. XP is *criterially frozen*.

(22) **Edge features:**

1. $[\odot EF \odot]$ on β can attract to Spec, β P any XP that is not criterially frozen.
2. Subsequently, XP can move out from Spec, β P.

Criterial features and edge features differ in two ways. Firstly, while criterial features are searching XPs with certain features (e.g. $[\odot F \odot]$ looks for XP goals bearing $[F]$), edge features do not look for specific features on their goal; edge features can move any

non-frozen constituent (as articulated by e.g. Georgi 2014, 2017). For example, while *wh*-movement and subject raising are driven by distinct criterial features [$\textcircled{\text{WH}}$] and [$\textcircled{\text{D}}$], respectively, the edge feature [$\textcircled{\text{EF}}$] can move any XP constituent, including *wh*-phrases and DPs. I assume that there is no superiority with respect to [$\textcircled{\text{EF}}$], i.e. if there are two eligible (i.e. non-frozen) goals, [$\textcircled{\text{EF}}$] can move either independently of their relative height.⁴ One further general intuition in the literature, which I will also follow here, is that every constituent that gets attracted by [$\textcircled{\text{EF}}$] must be affected later in the derivation; if it does not, the derivation crashes (Chomsky 1995; Fox and Pesetsky 2005; Heck and Müller 2000, 2007; Bošković 2007; Chomsky 2013). For example, while an [$\textcircled{\text{EF}}$] feature on an embedded C is able to attract either a *wh*-object or the VP containing it because [$\textcircled{\text{EF}}$] is not looking for goals carrying particular features and does not obey superiority, moving the VP to Spec,CP via [$\textcircled{\text{EF}}$] will lead to the crash of the derivation unless VP is attracted by some feature in the matrix clause. In contrast, moving the *wh*-object to Spec,CP via [$\textcircled{\text{EF}}$] does not lead to the crash of the derivation if a [$\textcircled{\text{WH}}$] probes attracts it in the matrix clause. I assume that the search of criterial features is constrained by Relativized Minimality (Rizzi 1990) and the A-over-A Condition (Chomsky 1964).

The second difference between the two feature types is that while an XP moved by a criterial feature [$\textcircled{\text{F}}$] to Spec, β P gets frozen criterially in Spec, β P (i.e. XP cannot move further), XPs moved by edge features are not frozen and can move to a higher position. Following (Rizzi 2004), I assume that sub-extraction from criterially frozen XPs is still possible; it is just XP that gets frozen.

For the distribution of criterial and edge features, I rely on a more general interpretation of the Borer-Chomsky Conjecture (Borer 1984; Chomsky 1995): Whether heads of a category β in a given language have [$\textcircled{\text{EF}}$] and/or some criterial features is lexically determined. The distribution of certain (re-)merge features is constrained by (23). This constraint is a more restrictive version of the *Categorical Merge hypothesis* by Newman

(2021, 2024), which says that the distribution of merge features is determined on the level of categories rather than the level of individual heads. Under the more restrictive constraint in (23), heads of a certain category must only align with respect to two kinds of (re-)merge features. On the one hand, heads of a certain category β must either all have or all lack $[\circ_{EF}\circ]$. For example, I will claim that English Ts and the P'urhepecha Cs uniformly have $[\circ_{EF}\circ]$ but Cs in English and German uniformly lack $[\circ_{EF}\circ]$. Previous work has typically handled edge features as some special features assigned in some special way (see e.g. the Phase Balance idea in Heck and Müller 2000). In contrast, I treat $[\circ_{EF}\circ]$ as a feature type whose distribution is not different from other movement features: It can be associated with any category and it is ordered relative to other features on heads (in the latter respect, my approach is closer to Georgi 2014, 2017). How $[\circ_{EF}\circ]$ features in different languages may differ in the number of searches they can do is discussed in Section 3.4.

- (23) Whether a functional category β in given language is associated with $[\circ_{EF}\circ]$ and/or $[\circ\alpha\circ]/[\bullet\alpha\bullet]$ such that β and α belong to the same *fseq* is the property of the particular category β (i.e. β heads will be uniform in this respect).

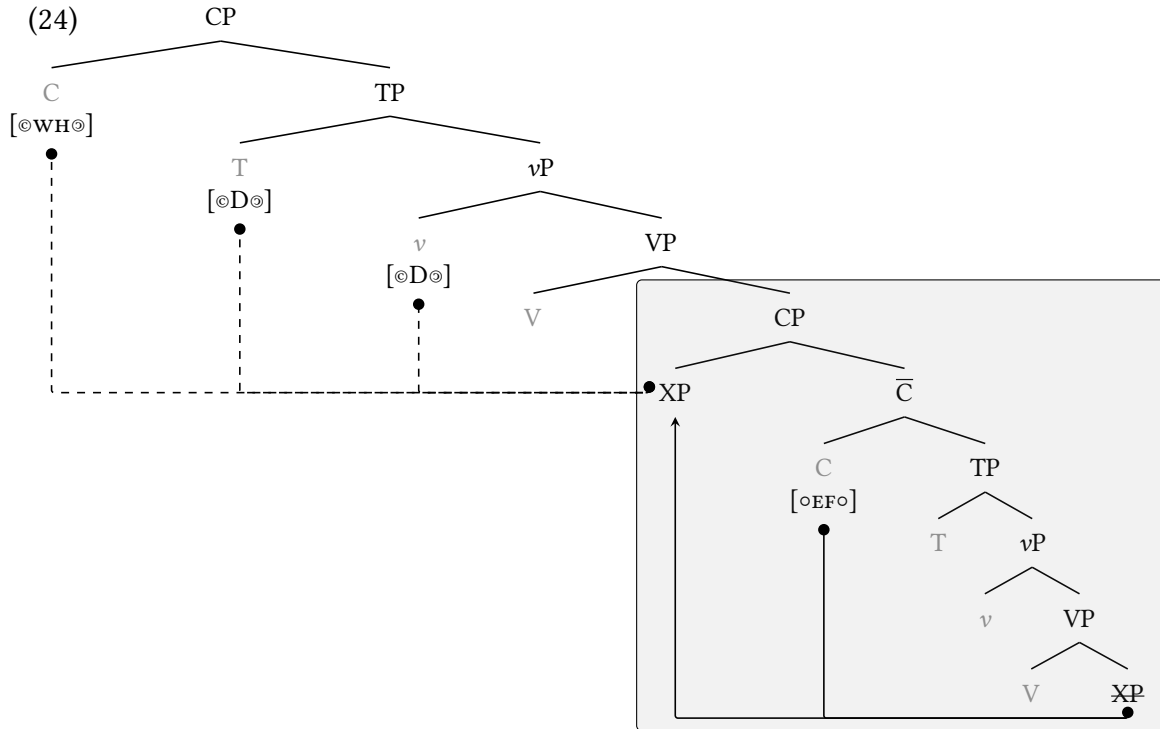
Under (23), there is a second class of features whose distribution is determined for entire categories: probes selecting for categories of the same *fseq* as the category of the head that the probe is located on. The first point where this is relevant is the building of extended projections. For example, *vs* uniformly have $[\bullet V\bullet]$ to select for a VP, Ts uniformly have $[\bullet v\bullet]$ to select for a *v*P, and Cs uniformly have $[\bullet T\bullet]$ to select for a TP; it cannot be the case that one C head has $[\bullet v\bullet]$ and selects for a *v*P and another has $[\bullet T\bullet]$ and for a TP. (Remember that the presence of a higher functional category implies the presence of all the lower ones.) At the same time, since (23) does not constrain what lexical categories like V can select for, Vs can differ in whether they select for a *v*P, a TP, or a CP. How (23) also constrains category-driven clausal movement is discussed in Section 4.4.

Beyond $[\circ_{EF}\circ]$ s and probes looking for categories from the *fseq* of their location, (23) does not constrain the distribution of probes. Thus, within a given language, Cs can differ

whether they carry [$\textcircled{WH}$], which moves a *wh*-phrase; and Ts can differ whether they carry [$\textcircled{D}$], which moves a DP subject.

3.2 How does successive cyclic movement work under PAFF?

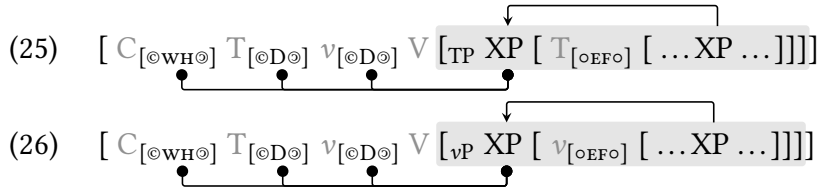
If a category β is associated with [$\textcircled{EF}$], PAFF permits successive cyclic movement via Spec, β P, that is, the edge status of such Spec, β P is preserved in the PAFF-based theory. In the schematic tree in (24), the embedded C has [$\textcircled{EF}$], and this [$\textcircled{EF}$] on C can reach the XP in the embedded VP along the well-formed C-T-*v*-V sequence (i.e. via clause-internal probing). Again, XP does not need to have any particular feature in order to be an eligible goal for [$\textcircled{EF}$] and [$\textcircled{EF}$] does not obey superiority; the only condition of successive cyclic movement is that it must feed some higher probe.



Upon moving to Spec,CP, XP in (24) c-commands *all* embedded heads and is c-commanded by *all* matrix heads. Since PAFF is defined in terms of the categories of c-commanding heads (rather than phrases), probes from matrix heads can uniformly

reach XP in Spec,CP, they do not need to consider any of the categories of the embedded clause. Thus, the $[\textcircled{\text{WH}}\textcircled{\text{D}}]$ feature on the matrix C can probe along the well-formed C-T- ν -V sequence; the raising probe $[\textcircled{\text{D}}\textcircled{\text{D}}]$ on the matrix T can probe along the well-formed T- ν -V sequence; and the raising probe $[\textcircled{\text{D}}\textcircled{\text{D}}]$ on the matrix ν can probe along the well-formed ν -V sequence. In other words, successive cyclicity breaks down cross-clausal movement into clause-internal and hence PAFF-complaint steps of probing. In general, successive cyclicity feeds *any* matrix probes without restrictions.

It does not matter what the category of the highest head of an extended projection is, as long as it has $[\textcircled{\text{EF}}\textcircled{\text{D}}]$, XPs moved by $[\textcircled{\text{EF}}\textcircled{\text{D}}]$ are accessible for any probe in the next lowest extended projection. In (25), the T head of the embedded TP bears $[\textcircled{\text{EF}}\textcircled{\text{D}}]$, in (26), the ν head of the embedded ν P bears $[\textcircled{\text{EF}}\textcircled{\text{D}}]$. Successive cyclicity works in the same way as it did with $[\textcircled{\text{EF}}\textcircled{\text{D}}]$ on C in (24): $[\textcircled{\text{EF}}\textcircled{\text{D}}]$ on T in (25) and on ν in (26) can move XP to a position where it c-commands all embedded heads and it is c-commanded by all matrix heads. Subsequently, any matrix probe can reach XP in Spec,TP in (25) and Spec, ν P (26).



In sum, my claim is that if the highest head β of a given β P has $[\textcircled{\text{EF}}\textcircled{\text{D}}]$, this β P exhibits *general transparency* or *no opacity*. That successive cyclicity makes XPs available for matrix probes irrespective of their height in the matrix clause implies that Williams Cycle effects can be obviated by successive cyclicity. For example, the $[\textcircled{\text{D}}\textcircled{\text{D}}]$ probes on T and ν could find XP in Spec,CP in (24), even though this is a violation of the Williams Cycle given that $C > T > \nu$ in *fseq*.

In the main text of this article, I do not discuss examples for Case and Agree. It can be easily seen that if the matrix heads C, T, or ν had Agree probes (e.g. $[\star\phi\star]$) in (18), (25), and (26), these Agree probes would work just like movement probes (e.g. $[\textcircled{\text{D}}\textcircled{\text{D}}]$ or

[$\odot_{WH}\odot$]): As long as some goal is below the highest head of the embedded clause, it is inaccessible, but as soon as the goal successive cyclically moves to the highest specifier of the embedded clause, matrix Agree probes can reach it along exclusively matrix heads, in a PAFF-complaint way. That this correctly predicts Williams Cycle obviation in Agree is briefly discussed in Footnote 20. Finally, Case assignment can also be re-analyzed as Agree (Chomsky 2000, 2001; Legate 2008; Poole 2024).

In the following two sub-sections, two examples will illustrate my claim that successive cyclicity obviates Williams Cycle effects: The English TP and the P'urhepecha CP.

3.3 Successive cyclic Williams Cycle-obviation in English TPs

English TPs are transparent to raising to the matrix Spec,TP subject position, as shown in (27) (Baltin 1995:p.224). Baltin (1995) notes that the quantifier floating of *all* indicates that subject raising to the matrix Spec,TP proceeds from embedded TPs via the embedded Spec,TP. (Subjects are assumed generated internal to *v*P.) In terms of the current framework, I take quantifier floating as an indicator of the presence of an [$\odot_{EF}\odot$] on the English T, which enables successive cyclic movement via Spec,TP. The successive cyclic raising of subjects is PAFF-compliant just as it was in (25): [$\odot_{EF}\odot$] probes along the singleton sequence of T, [$\odot_D\odot$] on the matrix T probes along the well-formed T-*v*-V sequence.

(27) $\downarrow$ ----- $\downarrow$
 [_{TP} They₁ seemed [_{TP} *all* ____ to ____ be friendly to us]]].

Since the English TP can be escaped successive cyclically, the PAFF-based theory predicts that any matrix probe can reach Spec,TP. That is, if there is a sufficiently low probe in the matrix clause, it should be able to attract XPs from English TPs even if this violates the Williams Cycle. This is borne out: *Raising to object* can proceed from TPs, as shown by (28) (modified version of Johnson 1991:p.595). That the DP *George* leaves the embedded TP headed by *to* is indicated by the fact that it precedes the *out* particle of the matrix verb *make*.⁵ Assuming that the particle *out* does not leave the matrix VP, that

the verb *makes* moves to *v*, and that *will* is a T head, *George* lands in a Spec, γ P such that $T > v > \gamma > V$ in *fseq*. Raising from TP to Spec, γ P is an obviation of the Williams Cycle as $T > \gamma$ in *fseq*. While (28) does not obey the Williams Cycle, it is correctly predicted to be grammatical by PAFF, as depicted in (29): The embedded subject is first moved to Spec,TP by [$\circ$ EF $\circ$] on T, then the [$\circ$ D $\circ$] probe on γ probes along the well-formed γ -V sequence.

(28) Mikey [_{TP} will [_{vP} make [_{γ P} George₁ [_{vP} out [_{TP} ₁ to be a liar]]]]].

(29) [T v γ [$\circ$ D $\circ$] V [_{TP} XP [T [$\circ$ EF $\circ$] [... XP ...]]]]]]]

Under (23) proposed in Section 3.1, all English Ts have [$\circ$ EF $\circ$]. Where finite and non-finite Ts differ is the presence of [$\circ$ D $\circ$]: I assume only finite English Ts have [$\circ$ D $\circ$].

3.4 Successive cyclicity via a single edge: Hyperraising from P'urhepecha CPs

Hyperraising is typically defined as syntactic movement from CPs to some argument position, typically to Spec,TP or Spec, v P. Among other languages, hyperraising to various argument positions has been argued for Mongolian (Fong 2019), Zulu (Halpert and Zeller 2015; Halpert 2019), Buryat (Bondarenko 2017a,b), Brazilian Portuguese (Nunes 2008), Nez Perce (Deal 2018), Japanese (Tanaka 2002; Yoon 2007), and Korean (Yoon 2007). Given that hyperraising refers to movement of a CP-internal subject to matrix Spec,TP, Spec, v P, or some other position below Spec,TP, attested cases of hyperraising involve Williams Cycle obviation under the standard *fseq* $C > T > v > V$ (e.g. Williams 2003; Poole 2022). The PAFF-based theory predicts that hyperraising always arises via successive cyclicity, here I exemplify the PAFF-based analysis of hyperraising using P'urhepecha CPs. My discussion relies on Erik Zyman's data and assumptions (Erik Zyman p.c., Zyman 2017, 2018).

P'urhepecha CPs introduced by the complementizer *eska* 'that' are transparent to three different movement types. The first movement type is *wh*-movement, which is

exemplified in (30) and (31): *wh*-phrases move from embedded CPs to a position in the matrix left-periphery, which I refer to as Spec,CP (Zyman 2017:p.4; Erik Zyman, p.c.).

- (30) ζ [_{CP} *Ambe*₁ *uetarincha-sin-Ø-ga=ni* [_{CP} $__$ ₁ *eska Emilia k'uanatsinta-a-Ø-ka* $__$ ₁]]?
 what need-HAB-PRS-INT=3pS that Emily buy-FUT-PRS-SBJV
 ‘What do they need Emily to buy?’ (interrogative mood)

- (31) ζ [_{CP} [*Naki takukat-eecha-ni*]₁ *xarara-sin-Ø-gi*
 which book-PL-ACC seem-HAB-PRS-INT
 [_{CP} $__$ ₁ *eska jorhenguwaririri-cha k'amaku-Ø-Ø-ka=sī* $__$ ₁ *ia*]]?
 that student-PL finish-PFV-PRS-SJ=3pS already
 ‘Which books does it seem that the students have already finished?’

Secondly, P'urhepecha CPs are transparent to subject raising. In (32), the embedded subject moves to a position preceding the matrix verb (Erik Zyman, p. c.). For simplicity, I will analyze the landing site of subject raising as Spec,TP.⁶

- (32) [_{TP} [*Ts'ima uaripeku-eecha*]₁ *xarara-sin-Ø-di=sī*
 those war-PL seem-HAB-PRS-IND3=3pS
 [_{CP} $__$ ₁ *eska* $__$ ₁ *k'amara-Ø-Ø-ka=sī* *ia*]].
 that end-PFV-PRS-SBJV=3pS already
 ‘Those wars seem to have ended already.’

Finally, P'urhepecha CPs are transparent to raising to object, as in (33) (Zyman 2017:p.4). That (33) involves genuine raising to the matrix clause is indicated by the fact that the accusative-marked embedded subject precedes the matrix PP ‘with all her heart’. Zyman (2017) analyses the landing site of raising to object as Spec,vP.

- (33) *Emilia ueka-sin-Ø-di* [_{vP} *Xumo-ni*₁ *mitsita-ni jingoni*
 Emily want-HAB-PRS-IND3 Xumo-ACC heart-ACC with
 [_{CP} $__$ ₁ *eska* $__$ ₁ *jaruata-a-Ø-ka* *pauani*]].
 that help-FUT-PRS-SBJV tomorrow
 ‘Emily wants Xumo with all her heart to help her tomorrow.’

As mentioned above, hyperraising from CP to Spec,TP and Spec,vP does not obey the Williams Cycle assuming that the functional sequence is C>T>v>V. Correspondingly, in the PAFF-based system, we expect that it is successive cyclic movement via Spec,CP

that obviates Williams Cycle effects. To demonstrate that this is indeed the case, I will use Zyman's "single edge" argument. Zyman (2017, 2018) proposes that since P'urhepecha CPs have a single edge position, every CP can be escaped by a single XP. In the framework of the current paper, the "single edge" phenomenon is captured by claiming that the $[\circ_{EF}\circ]$ feature on P'urhepecha C can probe only once. For example, the $[\circ_{EF}\circ]$ on the embedded C in (30) and (31) moves the *wh*-phrase, as it is schematically represented in (34).⁷

$$(34) \text{ [CP} \checkmark \text{C}_{[\circ_{WH}\circ]} \text{[TP} \times \text{T}_{[\circ_{D}\circ]} \text{[vP} \times \text{v}_{[\circ_{D}\circ]} \text{[VP V [CP } wh \text{ [C}_{[\circ_{EF}\circ]} \text{[TP DPT [vP v [VP V } wh \dots$$

The question arises what happens when the matrix C has $[\circ_{WH}\circ]$ *and* the matrix T/v has $[\circ_{D}\circ]$, as in (34). Since $[\circ_{EF}\circ]$ does not obey superiority and its goals do not have to bear specific features, the $[\circ_{EF}\circ]$ on an embedded C can choose to evacuate the embedded subject or an embedded *wh*-phrase, even when the *wh*-phrase is asymmetrically c-commanded by the subject. When $[\circ_{EF}\circ]$ chooses to move the embedded *wh*-object in (34), the embedded subject cannot be raised *in addition* to the *wh*-phrase via the embedded Spec,CP as $[\circ_{EF}\circ]$ disappears after moving the *wh*-object. Raising to the matrix Spec,TP or Spec,vP cannot happen in the absence of an intermediate stop in Spec,CP because the $[\circ_{D}\circ]$ probe on T cannot probe along the T-v-V-C sequence and the $[\circ_{D}\circ]$ on v cannot probe along the v-V-C sequence. These sequences are not PAFF-compliant because V and C are not adjacent in *fseq*. However, since hyperraising is optional in P'urhepecha (Zyman 2017, 2018), that is, $[\circ_{D}\circ]$ on T/v can remain unchecked, it does not lead to ungrammaticality if hyperraising does not happen in (34). In such cases, the only overtly moved phrase from the embedded clause is the *wh*-phrase, whereas the embedded subject remains *in-situ*.

Since $[\circ_{EF}\circ]$ is not search for specific features, it can choose to move the embedded subject, too; this on its own is not ungrammatical because the movement of the subject to Spec,CP feeds matrix $[\circ_{D}\circ]$ probes on T and v (i.e. the DPs moved by $[\circ_{EF}\circ]$ are later affected). However, ungrammaticality does arise if the only edge of a CP is used to evacuate the embedded subject *instead* of the *wh*-phrase. In (35) and (36) and (Erik Zyman, p.c.

and Zyman 2017:p.4, respectively), the embedded subjects are moved by the [$\circ_{EF\circ}$] on the embedded C to the embedded Spec,CP from which the matrix [$\circ_{D\circ}$] probes raise the embedded subjects to matrix Spec,TP and Spec,vP, respectively. Notice that these instances of hyperraising involve two clause-internal and therefore PAFF-compliant instances of probing; again, see the discussion around (24). However, in these sentences, no *wh*-phrases can escape the embedded CP in a PAFF-compliant way because the [$\circ_{EF\circ}$] on the embedded C cannot be used after moving the embedded subject. In the absence of the successive cyclic option in (35) and (36), the *wh*-probe on the matrix C cannot probe down along the ill-formed C-T-v-V-C-T-v-V sequence (problem: V and C are not adjacent in the *fseq*), so *wh*-movement is ungrammatical. Assuming *wh*-movement is an obligatory operation, that is, any unchecked [$\circ_{WH\circ}$] in the end of the derivation leads to ungrammaticality, not moving the *wh*-object also leads to ungrammaticality in (35) and (36). In sum, when the matrix C has [$\circ_{WH\circ}$] and the matrix T/v has [$\circ_{D\circ}$], [$\circ_{EF\circ}$] needs to choose to feed [$\circ_{WH\circ}$] rather than [$\circ_{D\circ}$] because unchecked *wh*-probes lead to a crash in the derivation, whereas unchecked raising probes do not (i.e. the former is obligatory, the latter is optional).

- (35) $\downarrow$ -----
 $[CP \times C_{[\circ_{WH\circ}]} [TP DPT_{[\circ_{D\circ}]} [vP v [VP V [CP DP [C_{[\circ_{EF\circ}]} [TP DP T [vP v [VP V *wh*...$
 $\uparrow$ -----
 $??[CP [Naki \text{ takukat-eecha-ni}]_1 \text{ xarara-sin-}\emptyset\text{-gi} [TP \text{ jorhenguwaririri-cha}_2$
which book-PL-ACC seem-HAB-PRS-INT student-PL
 $[CP \text{ } __2 \text{ eska } __2 \text{ k'amaku-}\emptyset\text{-}\emptyset\text{-ka=s}\ddot{i} \text{ } __1 \text{ ia}]]]?$
that finish-PFV-PRS-SJ=3pS already
Intended: 'Which books does it seem that the students have already finished?'

- (36) $\downarrow$ -----
 $[CP \times C_{[\circ_{WH\circ}]} [TP T [vP DP v_{[\circ_{D\circ}]} [VP V [CP DP [C_{[\circ_{EF\circ}]} [TP DP T [vP v [VP V *wh*...$
 $\uparrow$ -----
 $??[CP \text{ } \textit{iAmbe}_1 \text{ uetarincha-sin-}\emptyset\text{-ga} [vP \text{ Emilia-ni}_2$
what need-HAB-PRS-INT=pS Emily-ni
 $[CP \text{ } __2 \text{ eska } __2 \text{ k'uanatsinta-a-}\emptyset\text{-ka } __1]]]?$
that buy-FUT-PRS-SBJV
Intended: 'What do they need Emily to buy?' (interrogative mood)

Examples (35) and (36) also show that raising to subject and object involves genuine movement from the embedded CP; otherwise we would not expect any interaction with long *wh*-movement (Zyman 2017, 2018).

Importantly, I do not assume that [$\circ_{EF}\circ$] features in all languages move only one constituent; for example, Mongolian CPs are escaped successive cyclically (Fong 2019) and hyperraising from a CP can co-occur with scrambling another argument from the same CP (Jurkhaichin 2025). The "one edge" phenomenon was just used here as one possible way of identifying successive cyclicity (e.g. besides quantifier floating in Section 3.3). Whether a feature can probe once or multiple times has been a well-known difference between the movement features of the various languages and this has been typically encoded in the lexical properties of the heads of these languages. For example, while the [$\circ_{WH}\circ$] feature in English can move a single *wh*-phrase, the [$\circ_{WH}\circ$] feature in Bulgarian can move multiple *wh*-phrases (a.o. Rudin 1988; Richards 1997; Zompì 2024). The same difference between the [$\circ_{EF}\circ$] of the P'urhepecha C and the [$\circ_{EF}\circ$] of the Mongolian C provides further support for treating [$\circ_{EF}\circ$] features as lexically assigned, which are not different from "normal features" like [$\circ_{WH}\circ$] with respect to their distribution and cross-linguistic variability. (My notation does not differentiate one-shot and multi-shot probes.)

3.5 Interim summary

In this section, I proposed that successive cyclic movement obviates Williams Cycle effects for all operations. This leads to the status of *no opacity*: given that a language has a certain set of cross-clausal operations, and a given category β lexically has [$\circ_{EF}\circ$], successive cyclic movement will allow any of these operations from β Ps.

The analysis of P'urhepecha CPs can be extended to any other attested cases of hyper-raising (e.g. Tanaka 2002; Yoon 2007; Nunes 2008; Halpert and Zeller 2015; Bondarenko 2017a,b; Deal 2018; Halpert 2019). While these previous analyses did not necessarily make

use of successive cyclicity, the data seems compatible with stopping in Spec,CP.

It needs to be emphasized that the Williams Cycle obviating effect of successive cyclicity is only realized when the language has sufficiently low probes in the matrix clause. For example, the possibility of Williams Cycle obviation for English TPs would not be detectable if raising to a lower Spec, γ P object position did not exist in the language because all the other movement types out of TPs land in Spec,TP or higher, that is, they all obey the Williams Cycle. In a similar way, if v and T did not have a raising feature [$\textcircled{D}$] in P'urhepecha, we would not have been able to see that successive cyclicity obviates Williams Cycle effects for P'urhepecha CPs.

Finally, the presence/absence of edge features is lexically specified per category per language. Hence, that one category is associated with [$\textcircled{EF}$] (i.e. clauses headed by it can be escaped successive cyclically) does not mean that all other categories in the language are associated with [$\textcircled{EF}$]. For example, while I argued that English Ts have [$\textcircled{EF}$], which enables successive cyclic Williams Cycle obviation, English Cs will be argued to lack [$\textcircled{EF}$] in the next section. As we will see, this derives the lack of hyperraising from English CPs.

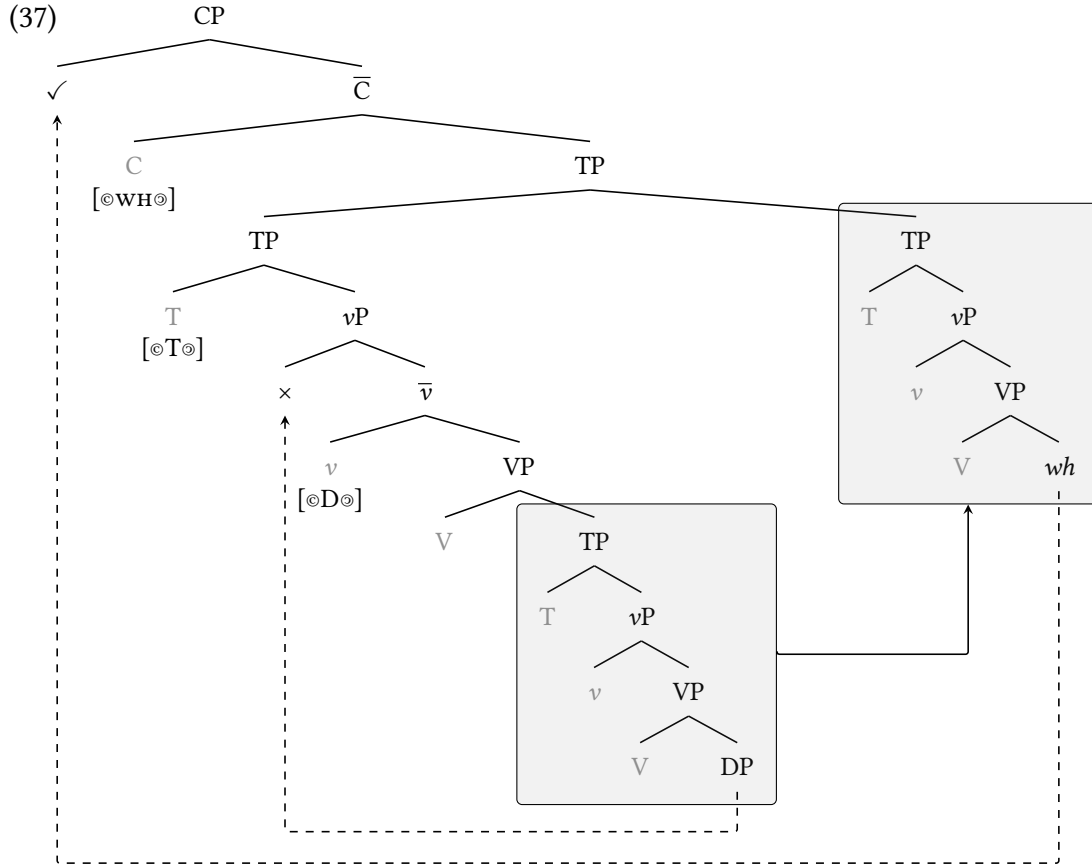
4 Clausal movement and Williams Cycle effects

4.1 Clausal movement as an alternative strategy

If the highest head of some constituent has no edge feature [$\textcircled{EF}$] lexically, successive cyclic movement cannot proceed from that constituent. As I will argue in this section, sub-extraction from such constituents is still possible provided that they move to an equal-sized matrix projection. However, while Williams Cycle effects were argued to be obviated by successive cyclicity, sub-extraction obeys the Williams Cycle if the transparency of the clause of origin is achieved by movement to an equal-sized matrix projection.⁸

The schematic derivation in (37) is used here to demonstrate how clausal movement feeds PAFF-complaint sub-extraction. Let us assume that Ts have no [$\textcircled{EF}$] in this hypothet-

ical language, no XP can use Spec,TP to escape the embedded TP, which is base-generated in the matrix VP. As discussed in Section 2, "one-fell-swoop" movement from TP to matrix projections is ruled out by PAFF, which is repeated in (38). For example, [$\textcircled{\text{D}}\textcircled{\text{O}}$] on the matrix v cannot probe along the ill-formed $v\text{-}\underline{\text{V-T}}\text{-}v\text{-}\text{V}$ sequence in a PAFF-compliant way because V and T are not adjacent in the *fseq* and $\text{T} > \text{V}$. For now, (37) illustrates the derivation with an embedded TP, which will be later generalized to any given embedded βP such that β has no [$\textcircled{\text{O}}\text{EF}\textcircled{\text{O}}$].



- (38) **Probe Along *Fseq*-Fragments (PAFF):** Within the domain dominated by the maximal projection of the head that the probe is located on, the sequence of category labels of the heads c-commanding the goal must correspond to a well-formed fragment of some functional sequence. (based on Egressy 2023)

I propose that PAFF-compliant sub-extraction can still proceed from the embedded TP in (37) if the *entire clause* undergoes movement to an equal-sized matrix projection,

that is, to a projection of the matrix TP. This movement step is driven by the $[\textcircled{\text{T}}]$ feature on the matrix T, and it is PAFF-compliant because the embedded TP is reached along the well-formed T- ν -V sequence. The movement of the embedded TP creates a setup for subsequent PAFF-compliant probing into that TP: Once the embedded TP has merged with a matrix TP projection, the sequence of embedded heads finds a continuation in the matrix clause. For example, $[\textcircled{\text{WH}}]$ on the matrix C can reach XP along the well-formed C-T- ν -V sequence. Although C is a head in the matrix extended projection, whereas T, ν , and V are heads in the embedded extended projection, their C-T- ν -V sequence is a well-formed fragment of the *fseq* $\text{C} > \text{T} > \nu > \text{V}$. That is, probing along this sequence is allowed under PAFF.

This clausal movement strategy of sub-extraction derives Williams Cycle effects: Embedded clauses are *in-situ* and hence opaque before the matrix head corresponding to the category of the embedded clause is merged in and provides a landing site for the embedded clause. In (37), the embedded TP is opaque to movement to the matrix Spec, ν P because this position is too low in the matrix clause. As discussed above, $[\textcircled{\text{D}}]$ on ν probes too early, before the movement of the embedded TP. Since Ts in this hypothetical language have no $[\textcircled{\text{EF}}]$, the *in-situ* TP is opaque to successive cyclic sub-extraction, too. Generalizing what we see in (37) with an embedded TP, embedded β P are opaque before but transparent after $[\textcircled{\beta}]$ on the matrix β moves them. That is, if there is a probe on γ , it cannot search into β P such that $\beta > \gamma$ in the *fseq* because β P has not merged yet with the matrix β P.

In sum, my claim in this section is that Williams Cycle-style selective opacity arises when sub-extraction from a clause is achieved via clausal movement rather than successive cyclicity. In terms of features, when β s can have no $[\textcircled{\text{EF}}]$ in a given language, sub-extraction can still proceed from β P if the β category is associated with $[\textcircled{\beta}]$ in the lexicon. However, then the sub-extraction pattern from such β P obeys the Williams Cycle because it is only possible after the movement of β P to an equal-sized matrix projection. In Section 4.2, I bring an example for this behavior: German CPs are opaque *in-situ* but

transparent upon extraposition to a matrix CP projection. Section 4.3 discusses what determines whether an embedded β P extraposed to a matrix β P projection is transparent to movement to the matrix Spec, β P, that is, whether sub-extraction can land in the matrix projection where extraposition lands. Section 4.4 argues that there is a good reason to believe that the category-driven movement of embedded clauses (i.e. β Ps moved by [$\textcircled{\text{c}}\beta\textcircled{\text{c}}$]s) must happen to an equal-sized matrix projections universally and independently of sub-extraction (e.g. CPs move to CP, TP_s move to TP). In Section 4.5, I use Dutch CPs to demonstrate that while embedded clauses can be moved to non-equal-sized matrix projections when the movement is not category-driven (i.e. β Ps not moved by [$\textcircled{\text{c}}\beta\textcircled{\text{c}}$]), such clausal movement does not sub-extraction, as predicted by PAFF. Section 4.6 suggests a PAFF-compatible analysis for the Williams Cycle effects in the opacity of English CPs.

4.2 German CPs are transparent only upon extraposition

In this section, I demonstrate that German CPs exhibit the behavior described in Section 4.1: They are only transparent to sub-extraction after they move to a matrix CP projection.

German clauses introduced by *dass* ‘that’ are standardly analyzed as CPs (Keine 2016, a.m.o.). German CPs like the ones in (39) are verb final and the CP objects of larger verb-final CPs are generated in the same pre-verbal position as normal objects. At least some CPs can remain in their original pre-verbal position, as shown by (39) (Müller 1999:p.17).⁹¹⁰

- (39) ...*dass er* [_{CP} *dass der Fritz Claudia liebt*] *gesagt hat*.
 that he that the Fritz.NOM Claudia.ACC love.3SG say.PTCP have.3SG
 ‘that he said that Fritz loves Claudia.’

At the same time, CPs typically extrapose to the right, to a post-verbal position, as in (40) (Müller 1999:p.17). I do not discuss here what decides whether extraposition is obligatory or optional (for further details, see Müller 1999).

- (40) *dass er* ₁ *gesagt hat* [_{CP} *dass der Fritz Claudia liebt*]₁.
 that he say.PTCP have.3SG that the Fritz.NOM Claudia.ACC love-3SG
 ‘that he said that Fritz loves Claudia.’

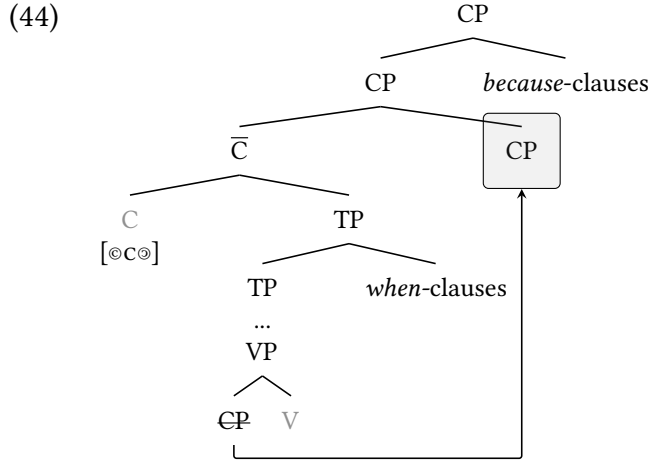
To identify the landing site of German CPs when they extrapose, I rely on the relative height of TP-adjuncts and CP-adjuncts. Since *als*-clauses, ‘when’-clauses, are temporal modifiers, I handle them as TP-adjuncts. As shown by (41), such TP-adjuncts can precede extraposed CPs in the right periphery (see also Müller 2020).

- (41) *Peter hat* _{—1} *gesagt* [*als er Maria gesehen ha-t*]
 Peter.NOM have.3SG say.PTCP when he.NOM Maria.ACC see.PTCP have.3SG
 [_{CP} *dass er jetzt losgehen muss*]₁.
 that he.NOM now away.go.INF must.3SG
 ‘When he saw Maria, Peter said he must leave now.’

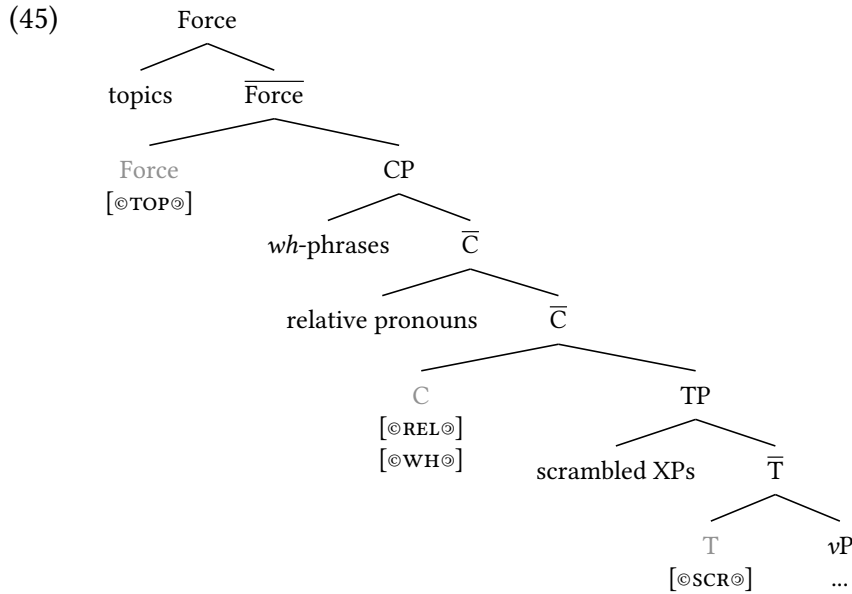
Weil-clauses, ‘because’-clauses, are handled here as CP-adjuncts. Unlike TP-adjuncts, CP-adjuncts cannot precede but must follow extraposed CPs, as shown by (42) and (43) (Müller 2020).

- (42) *Peter hat* _{—1} *gesagt* [_{CP} *dass er jetzt losgehen muss*]₁
 Peter.NOM have.3SG say.PTCP that he.NOM now away.go.INF must.3SG
 [*weil er Maria gesehen hat*].
 because he.NOM Maria.ACC see.PTCP have.3SG
 ‘Because he saw Maria, Peter said that he must leave now.’
- (43) ?? *Peter hat* _{—1} *gesagt* [*weil er Maria gesehen*
 Peter.NOM have.3SG say.PTCP because he.NOM Maria.ACC see.PTCP
hat] [_{CP} *dass er jetzt losgehen muss*]₁.
 have.3SG that he.NOM now away.go.INF must.3SG
 Intended: ‘Because he saw Maria, Peter said that he must leave now.’

German CPs extrapose to a position above TP-adjuncts but below CP-adjuncts, as depicted in (44). From this relative position and assuming that within a given phrase, movement/extraposition lands below the adjuncts of the phrase (Hunter 2010; 2015, a.o.), I infer that it is the C category in German that is associated with the [_ⓈC_Ⓢ] extraposition feature and *because*-clauses adjoin to the matrix CP above the extraposition site of embedded CPs.¹¹ In Section 4.4, I propose that category-driven clausal movement universally happens to an equal-sized matrix projection.



In this article, I distinguish three landing heights in the German left-periphery, which are based on the assumptions of Keine (2016, 2019, 2020) and are depicted in (44).



The highest position is the position of topics, which is analyzed as Spec,ForceP and the probe driving topicalization is [©TOP©] on Force. The topic position is also known as the highest specifier of verb-second clauses, in which the verb moves up to Force. The second highest specifier is Spec,CP, the landing site of *wh*-movement and the movement of relative pronouns, whose probes on C are [©WH©] and [©REL©], respectively. Here, I focus on *wh*-movement landing in the highest specifier of verb-final CPs, which correspond

to embedded questions. Finally, scrambling lands in Spec,TP and its probe is [_QSCR_Q] on T.¹² This subsection discusses the possibility of the subextraction of topics and scrambled phrases from German CPs, the next subsection discusses the possibility of subextraction of *wh*-phrases and relative pronouns.

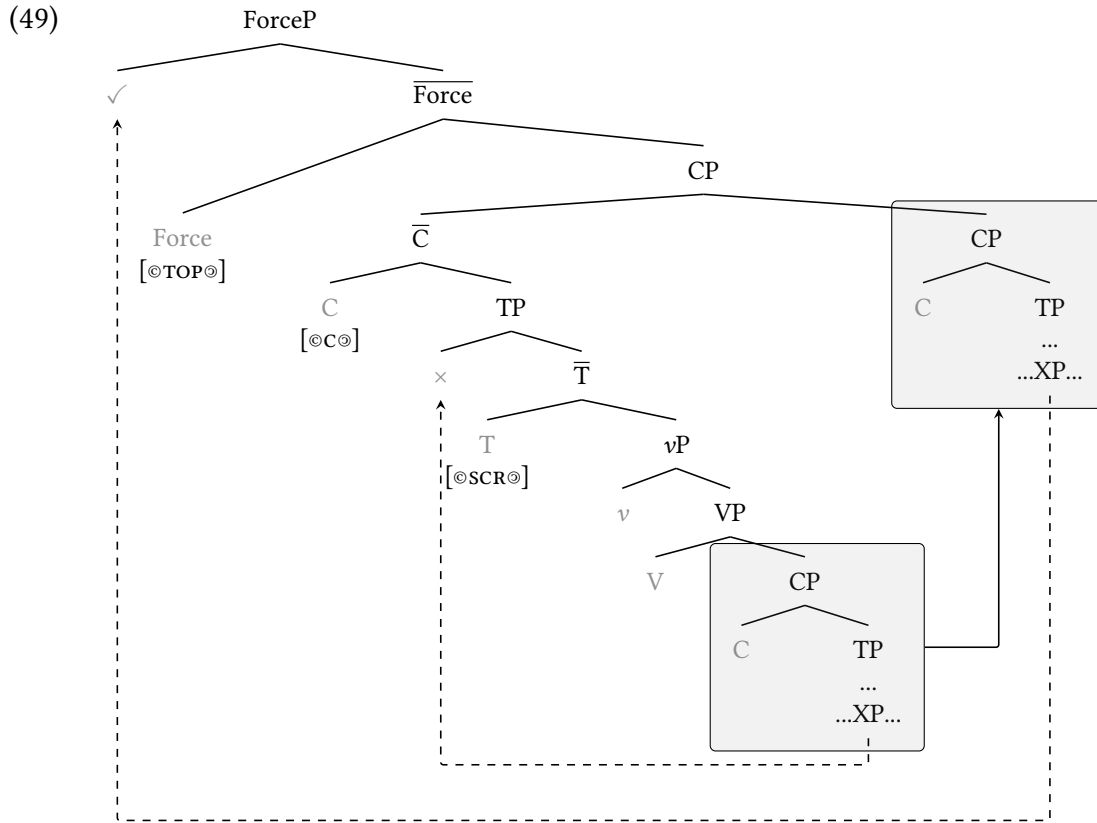
Crucially, extraposition influences the opacity of CPs: Topicalization, which is assumed to land in Spec,ForceP, can proceed from extraposed CPs, as in (46) (Müller 1999:p.38). In contrast, *in situ* CPs are opaque to topicalization, as in (47) (Müller 1999:p.38).

- (46) [*Die Maria*]₁ *habe* *ich* __₂ *gedacht* [*CP* *dass der Fritz* __₁ *liebt*]₂.
 the Maria.ACC have.1SG I think.PTCP that the Fritz.NOM love.3SG
 ‘Maria, I thought that Fritz loves.’
- (47) * [*Die Maria*]₁ *habe* *ich* [*CP* *dass der Fritz* __₁ *liebt*] *gedacht*.
 the Maria.ACC have.1SG I that the Fritz.NOM love.3SG think.PTCP
 Intended: ‘Maria, I thought that Fritz loves.’

In other words, German CPs provide an actual, real world example of the PAFF-based account of sub-extraction-via-clausal-movement, which was laid out with schematic derivations in the previous subsection: Topicalization out of a German CP requires that the CP extraposes to a matrix CP. In formal terms, I propose that German Cs are not like heads like P'urhepecha Cs or English Ts in that they are never associated with $[\circ_{\text{EF}}\circ]$.¹³ Consequently, the opacity of German CPs is in line with the theory introduced in Section 4.1. (48) depicts the opacity of an *in situ* CP. (In my diagrams, I do not depict any German projections as head-final for easier readability.) When the embedded CP occurs in its original VP-internal matrix position, no sub-extraction can proceed from it: Without successive cyclicity via Spec,CP, the embedded object cannot be reached by the matrix Force's $[\circ_{\text{TOP}}\circ]$ along the Force-C-T-v-V-C-T-v-V sequence (problem: $V \not\prec C$ in *fseq*).

- (48) [Force_{[[⊙] TOP [⊙]]} C T v V [CP C T v V XP]]

In contrast, PAFF-compliant sub-extraction is possible if the embedded CP has moved to the matrix CP, as in (49): the [$\textcircled{\text{TOP}}$] feature on the matrix Force can probe along the well-formed Force-C-T- ν -V sequence and reach the XP to be topicalized in the embedded VP. That is, it is correctly derived that topicalization from extraposed CPs is grammatical, as in (46). (That the embedded CP in German indeed extraposes to matrix CP was demonstrated above.)



The argumentation in Section 4.1 predicts that since German CPs are escaped via clausal movement (rather than successive cyclicity), the sub-extraction pattern from German CPs follows the Williams Cycle. This is borne out once we consider scrambling. Following Keine (2020), I assume that German scrambling lands in Spec,TP. For example, the object of the infinitival clause in (50) is moved to the Spec,TP position of the finite *weil*-clause, where it precedes the subject *jemand* but follows the C head *weil* (Keine 2020:p.213).

- (50) *weil* [_{TP} [_{den Wagen}]₁ *jemand* [_{vP} ___₁ *zu reparieren*] *versucht ha-t*]
 because the car.ACC someone.NOM to repair.INF try.PTCP have-3SG
 ‘because someone tried to repair the car’

As exemplified by (51), scrambling is impossible from CPs, even from extraposed CPs (Müller 2014a:p.117). Given that scrambling lands in Spec,TP in German, the ban on scrambling from CPs can be analyzed as a Williams Cycle effect because C>T in *fseq*. As illustrated in (49), the impossibility of scrambling from extraposed/non-extraposed CPs is accounted for by the PAFF-based theory: While topicalization landed in Spec,ForceP, which is higher than the CP-internal extraposition-site of CPs, scrambling lands in a position, Spec,TP, which is below the extraposition-site. More specifically, the embedded CP is *in situ* in the matrix VP at the point in the derivation when the [_{SCR}] probe on the matrix T is searching for a goal, so its probing proceeds along the ill-formed T-*v*-V-C-T-*v*-V sequence. This is not PAFF-compliant because C is not above V in *fseq*.

- (51) *dass*[_{TP} [_{*das Buch}]₁ *Karl* ___₂ *glaubt*]
 that the book Karl.NOM think.3SG
 [_{CP} *dass* [_{TP} [_{*das Buch}]₁ *keiner* ___₁ *lies-t*]₂]
 that the book.ACC no.one.NOM read-3SG
 ‘that Karl thinks that no one reads the book.’

4.3 Extraposition and subextraction within the same projection

Topicalization and scrambling land in a higher or lower projection, respectively, than extraposition site of German embedded CPs. The sub-extraction of *wh*-phrases and relative pronouns requires special attention because these land Spec,CP, that is, within the same projection as the extraposed CPs from which the sub-extraction tries to proceed.

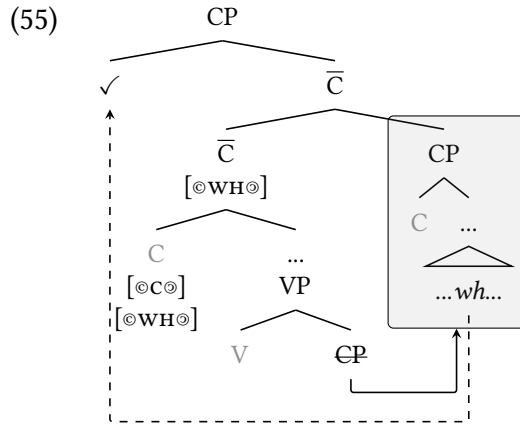
Wh-movement shows the same subextraction pattern as topicalization: While extraposed CPs like (52) are transparent to *wh*-movement (Müller 1999:p.18), *in situ* (pre-verbal) CPs like (53) are opaque to *wh*-movement (Müller 1999:p.18). (There is some cross-dialectal variation in the acceptability of (52), which is addressed at the end of this section.)

- (52) *Ich weiß nicht* [_{CP} *wen₁ er* ₂ *gesagt hat* [_{CP} *dass Fritz* ₁ *liebt*]₂].
 I knew.1SG not who.ACC he said have.3SG that Fritz love.3SG
 ‘I did not know who he said that Fritz loves.’
- (53) * *Ich weiß nicht* [_{CP} *wen₁ er* [_{CP} *dass Fritz* ₁ *liebt*] *gesagt hat*].
 I knew.1SG not who.ACC he that Fritz love.3SG said have.3SG
 Intended: ‘I did not know who he said that Fritz loves.’

As already discussed in the previous section, given that German Cs do not have [_{oEFo}], the opacity of *in-situ* CPs falls under the PAFF-based theory. As shown in (54), [_{oWHo}] on the matrix C cannot probe along the ill-formed C-T-*v*-V-C-T-*v*-V sequence because V and C are not adjacent in *fseq*.

- (54) [_C[_{oWHo}] T *v* V [_{CP} C T *v* V XP]]


Since both CP-extraposition and *wh*-movement land within the CP projection of the matrix clause, both [_{oCo}] and [_{oWHo}] occur on C in German. If [_{oCo}] is extrinsically ordered before [_{oWHo}] on C (on ordering, see Georgi 2014, 2017), PAFF correctly predicts CP-extraposition can feed *wh*-movement, as depicted in (55). First [_{oCo}] makes the embedded CP merge with a matrix CP projection. Since [_{oWHo}] is the second probe on [_{oCo}] and it is still active/unchecked after the CP-extraposition, it projects within the matrix CP and it probes into the embedded CP from a c-commanding non-minimal CP-projection, which is here marked with $\overline{C}$. Under PAFF, the domain we have to consider for heads c-commanding the goal is the maximal CP projection in the matrix clause. Within this domain, the sequence of heads c-commanding the *wh*-phrase in the embedded VP includes exclusively embedded heads: C-T-*v*-V. Hence, [_{oWHo}] can reach the *wh*-phrase in the embedded CP in a PAFF-compliant way. In other words, (55) shows that whenever probing does not happen from a head but from a higher projection, here from a projection marked as $\overline{C}$, the category of the location of the probe, here the matrix C, is not considered under PAFF. This contrasts with cases discussed in the previous section, which probing from a head and hence the highest head to be considered was always the location of the probe.



The landing site for relativization in German is the same as for *wh*-movement, yet the two movement types have different locality restrictions (Keine 2016, 2020). As demonstrated in (56), German relativization lands in the Spec,CP position of head-final finite clauses and it can proceed from infinitival clauses (Keine 2020:p.214). Therefore, I assume that the [©REL©] probe is on the C head.

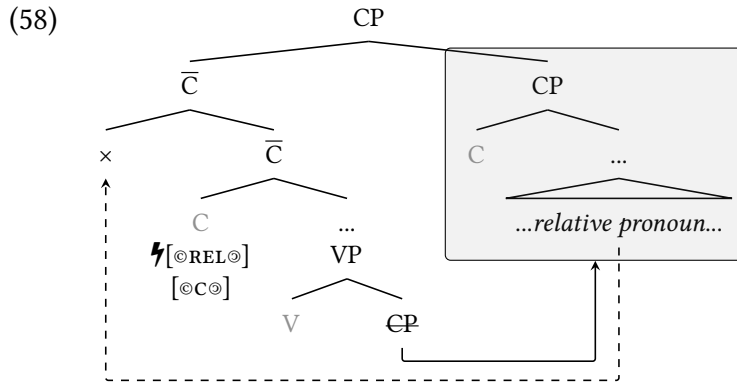
- (56) *der Wagen* [_{CP} *den₁* *er* [_{TP} *___₁* *zu reparieren*] *abgelehnt ha-t*]
 the car which.ACC he to repair.INF refuse.PTCP have-3SG
 ‘the car which he refused to repair’

However, unlike *wh*-movement in (52), relativization cannot proceed from German CPs (Salzmann 2006), as exemplified by (57) (Keine 2020:p.219). As for scrambling, it does not matter whether the embedded CP is extraposed or not.

- (57) ?? *der Junge* [_{CP} *den₁* *sie* *___₂* *glaubt*
 the boy who.ACC she believe.3SG
 [_{CP} *dass der Peter* *___₁* *getroffen hat*]₂]
 that the Peter meet.PTCP have.3SG
 Intended: ‘The boy that she believes that Peter has met.’

The extrinsic ordering of syntactic probes (Georgi 2014; 2017; a.o.) helps us to capture that German CPs are transparent to *wh*-movement but opaque to relativization. I propose that while the [©WH©] probe is ordered after the [©C©] probe, as it was illustrated in (55), the relative probe [©REL©] is ordered before the [©C©] probe, as in (58). Consequently,

while *wh*-movement in (55) emerges as a result of *feeding*, [$\textcircled{\text{C}}$] feeds [$\textcircled{\text{WH}}$], (58) can be characterized *counterfeeding*, the movement of the CP comes too late to feed [$\textcircled{\text{REL}}$]. More precisely, [$\textcircled{\text{REL}}$] tries to probe into the *in situ* rather than the extraposed CP, which is impossible along the ill-formed C-T-v-V-C-T-v-V sequence (problem: $\text{V} \not\geq \text{C}$ in *fseq*).



Various Western Germanic dialects/languages exhibit variation in the possibility of *wh*-movement and relative movement from extraposed CPs. This is captured by the various feature orderings on the Cs of these dialects in (59). I refer to the Standard/High German dialect characterized above as Type A. As discussed above, this dialect allows *wh*-movement but not relativization from CPs, which is derived if we order [$\textcircled{\text{REL}}$] before but [$\textcircled{\text{WH}}$] after [$\textcircled{\text{C}}$] on Cs; $\text{X} > \text{Y}$ means "X before Y" in (59). Certain Standard/High German speakers accept neither *wh*-movement nor relativization to Spec,CP from CPs (Keine 2020). (59) refers to these speakers as Type B. For these speakers, [$\textcircled{\text{WH}}$] and [$\textcircled{\text{REL}}$] are ordered on C before [$\textcircled{\text{C}}$], that is, not just [$\textcircled{\text{REL}}$] but also [$\textcircled{\text{WH}}$] tries and fails to probe into embedded CPs *before* the CP is extraposed by [$\textcircled{\text{C}}$]. Finally, speakers of Western Germanic languages/dialects like Bavarian and Dutch accept both *wh*-movement and relativization from extraposed CPs (Bayer 1984; Boef 2013). This pattern can be captured by ordering [$\textcircled{\text{REL}}$] and [$\textcircled{\text{WH}}$] after [$\textcircled{\text{C}}$] on the C head. That is, not only [$\textcircled{\text{WH}}$] but also [$\textcircled{\text{REL}}$] is fed by the extraposition of CPs. I am not aware of any Western Germanic dialects exhibiting the opposite of Type A High German, that is, allowing relative movement but

not *wh*-movement from CPs (for further discussion, see Bader and Koukouloti 2024).

	Feature ordering on C	Rel.mov.	Wh-mov.	Western Germanic dial.
(59)	$[\textcircled{\text{WH}}], [\textcircled{\text{REL}}] > [\textcircled{\text{C}}]$	×	✓	Type A High German
	$[\textcircled{\text{REL}}] > [\textcircled{\text{C}}] > [\textcircled{\text{WH}}]$	×	×	Type B High German
	$[\textcircled{\text{C}}] > [\textcircled{\text{WH}}], [\textcircled{\text{REL}}]$	✓	✓	Bavarian, Dutch

4.4 Move to same category

In the derivations in Sections 4.1, 4.2, and 4.3, it was essential for my proposal that the category-driven movement of embedded clauses landed at an equal-sized matrix projection. In Section 4.2, I brought evidence showing that in German, embedded CPs indeed move to a matrix CP projection. In this section, I argue that the landing site of German CPs was not accidental but follows the cross-linguistic pattern described by the theorem in (60). For example, according to (60), it holds cross-linguistically that the category *v* can be associated with $[\textcircled{\text{v}}]$ moving *v*P-sized clauses but not with $[\textcircled{\text{C}}]$ or $[\textcircled{\text{T}}]$ moving CP-sized and TP-sized clauses, respectively. In a similar way, T can be associated with $[\textcircled{\text{T}}]$ moving TP-sized clauses but not with $[\textcircled{\text{C}}]$ ¹⁴ or $[\textcircled{\text{v}}]$ moving CP-sized and *v*P-sized clauses, respectively. Finally, C can be associated with $[\textcircled{\text{C}}]$ moving CP-sized clauses but not $[\textcircled{\text{T}}]$ or $[\textcircled{\text{v}}]$ moving TP-sized and *v*P-sized clauses, respectively. In general, (60) describes the universal distribution of features driving category-based clausal movement.

- (60) **Move to same category:** If the extended projection topped by α P is moved via an $[\textcircled{\alpha}]$ feature on a functional head β to a higher extended projection and α and β belong to the same *fseq*, it must be the case that $\alpha = \beta$

(60) constrains only category-based movement, when it is the category label of some β P that makes it an eligible goal, when a $[\textcircled{\beta}]$ is capable of moving any β P because all that matters is the β -ness of the constituent (e.g. $[\textcircled{\text{C}}]$ moves CPs for their C-ness). What (60) does not constrain includes cases when an extended projection moves because it has a special status in the information structure. For example, $[\textcircled{\text{FOC}}]$ or $[\textcircled{\text{TOP}}]$ drive non-category-based movement: they move only those CPs that are foci or topics, and

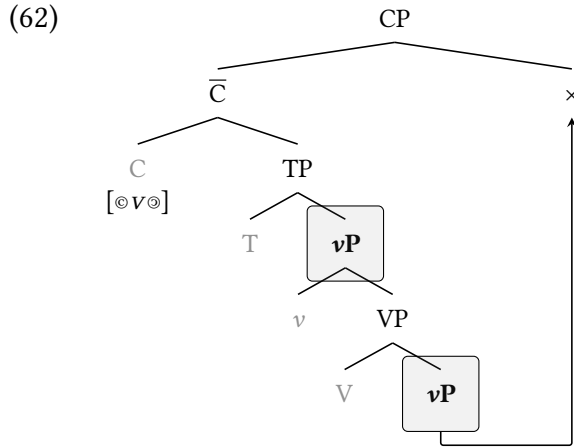
they move topic/focus constituents of other categories (e.g. DPs or TPs), too (see e.g. the topicalization of German CPs in Müller 1999; for an attempt to remove the category/non-category contrast, see Cable 2007, 2010). For an example from Dutch, see Section 4.5.

The theorem in (60) follows from two components. On the one hand, I claim that there is a principle saying that clauses cannot be moved to matrix projections whose *fseq*-index is lower (i.e. to smaller matrix projections). On the other, I will argue that the A-over-A Condition (Chomsky 1964) derives that embedded clauses cannot be moved to matrix projections whose *fseq*-index is higher (i.e. to larger matrix projections).

Firstly, I claim that some β cannot bear $[\circ\alpha\circ]$ if $\alpha > \beta$ in *fseq* (e.g. ν cannot bear $[\circ C\circ]$). Here I rely on the general architecture of extended projections. According to Grimshaw (2000), a functional sequence is an ordered sequence of functional heads with a lexical core/lowest element. For example, C, T, ν are functional heads; V is a lexical head. Extended projections are built up as a sequence of local category selections: For example, ν selects for VP via $[\bullet V\bullet]$; T selects for νP via $[\bullet \nu\bullet]$; and C selects for TP via $[\bullet T\bullet]$. It is reasonable to propose the universal principle in (61): No functional head can select for a higher *fseq*-category (e.g. T cannot bear $[\circ C\circ]/[\bullet C\bullet]$, ν cannot bear $[\circ T\circ]/[\bullet T\bullet]$). Notice that (61) does not hold for V because it is lexical category, so V can select for any size of a complement clause (e.g. νPs , TPs, and CPs). The general intuition behind (61) is that it holds not only for external but also for internal merge that heads can only select for those categories in the same *fseq* that are present in the previously built part of extended projection.¹⁵ For example, while building a TP constituent, T, ν , and V are already present in the lower portion of the clause, whereas C is not. In short, (61) formalizes and generalizes the idea by Starke (2001) and Adger (2003) that if two functional categories of the same *fseq* merge, the category with a higher *fseq*-index projects.

- (61) **Do not select for larger:** A functional head γ cannot have $[\circ\beta\circ]/[\bullet\beta\bullet]$ if $\beta > \gamma$ in some functional sequence.

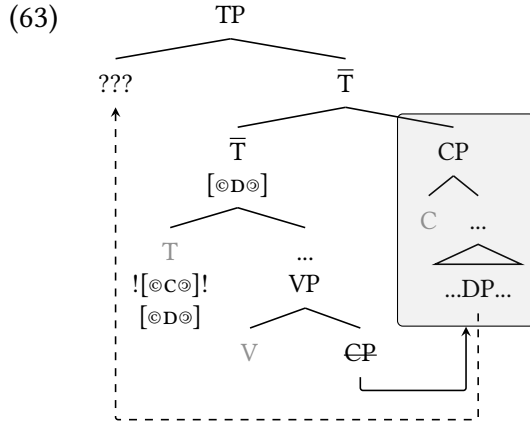
The second component of the theorem in (60) is that the movement of smaller clauses to larger matrix projections is impossible. This follows from the A-over-A Condition (Chomsky 1964):¹⁶ As proposed in Section 2, the presence of higher categories in the extended projection implies the clause-internal presence of all the lower ones (e.g. no clause can have a CP without a TP and ν P). Therefore, if a matrix head β has $[\ominus\gamma\ominus]$ such that $\beta > \gamma$ in *fseq*, this feature finds the clause-internal γ P first and hence blocks the movement of embedded γ P from the complement of V. This is illustrated by in (62): The matrix ν P is closer to the $[\ominus\nu\ominus]$ feature of the matrix C, so the lower ν P cannot move to the matrix CP.



In sum, since β cannot have $[\ominus\alpha\ominus]$ such that $\alpha > \beta$ in *fseq* under the principle in (61) and $[\ominus\gamma\ominus]$ features on β s such that $\beta > \gamma$ in *fseq* will fail to move embedded γ P clauses, (60) is derived.

The relevance of (60) comes from configurations like (63). (60) does not allow the embedded CP to be moved to the matrix TP and be directly followed by subject raising. This is essential because otherwise, $[\ominus D \ominus]$ on the projection marked here as $\bar{T}$ could probe along the well-formed C-T- ν -V sequence of embedded heads and reach embedded constituents like the embedded subject. (Just like German CPs were argued to be transparent to PAFF-complaint *wh*-movement to matrix Spec,CP immediately after their extraposition

to a matrix CP projection in Section 4.3.) In other words, the theory would lose a lot from its restrictiveness because Williams Cycle obviation would be possible even without successive cyclic movement.



Similarly to edge features, I assume that the features driving clausal movement are associated with individual categories in individual languages. The constraint in (64), which was introduced in Section 3.1, proposes a category-level assignment for not only edge features but also probes that are searching for a category from the *fseq* of the head that the probe is located on. That is, if a given category β in a given language is associated with $[@\beta@]$, all heads of the β category have $[@\beta@]$.

- (64) Whether a functional category β in given language is associated with $[@\beta@]$ and/or $[@\alpha@]/[\bullet\alpha\bullet]$ such that β and α belong to the same *fseq* is the property of the entire category β (i.e. β heads will be uniform in this respect).

4.5 Movement to non-equal-sized projections fails to feed sub-extraction

In this section, I use data from Dutch to show that while the non-category-based movement of clausal projections can sometimes happen to non-equal sized clausal projections, this does not facilitate sub-extraction to higher projections. As demonstrated below, this restriction follows from the PAFF-based theory.

Dutch is a closely related language to German, so for simplicity, I assume that Dutch clauses have the same functional hierarchy as German clauses (Force > C > T > v > V). Just like

German CPs headed by *dass* ‘that’, Dutch CPs headed by *dat* ‘that’ are verb-final and can be extraposed. In the same way as in German, the extraposition can happen to a position above TP-adjuncts, as in (65), but must happen below CP-adjuncts, as in (66). Therefore, I assume that similar to German, it is the category C in Dutch that is associated with extraposition feature [_{EC}]. In short, the arrangement proposed for German CPs in (44) also applies for Dutch CPs.

- (65) *Jan had nooit —₁ gedacht [toen hij in de stad woonde] [CP*
Jan have.3SG.PST never think.PTCP when he in the city live.3SG.PST
dat hij een auto zou kopen]₁.
that he a car should.3SG buy.INF
 ‘When he was living in the city, Jan never thought that he should buy a car.’

- (66) *Jan had nooit —₁ gedacht { ?? omdat hij geen rijbewijs*
Jan have.3SG.PST never think.PTCP because he no drivers.license
heeft} [CP dat hij een auto zou kopen]₁ { omdat hij geen
have.3SG that he a car should.3SG buy.INF because he no
rijbewijs heeft}.
drivers.license have.3SG
 ‘Because he had no drivers license, Jan never thought that he should buy a car.’

As illustrated by (67), Dutch allows the movement of embedded CPs to the middle field of the matrix clause (Barbiers 2000; Van Urk and Richards 2015). That the CP has left its VP-internal position is shown by the fact that it is separated from the matrix verb by *nooit* ‘never’.

- (67) *Jan had [CP dat hij DAT zou kopen]₁ nooit —₁ gedacht].*
Jan have.3SG.PST that he that should.3SG buy.INF never think.PTCP
 ‘Jan did not think that he should buy THAT.’

Importantly, preposed clauses land lower than C in the matrix clause: As shown by (68), the preposed CP lands below the non-topicalized matrix subject. In the Dutch middle field, the highest position of non-topicalized subjects is Spec,TP (e.g. Broekhuis and Corver 2016). For simplicity, I will assume that preposed clauses land in Spec,vP and their movement is

driven by a scrambling [$\textcircled{\text{SCR}}$] probe on v .¹⁷ Barbiers (2000) claims that preposing has an effect on information structure: CPs moved to this position are necessarily interpreted as factives. That is, factivity is encoded here by the position for the embedded clause in the matrix clause rather than the internal structure of the embedded clause (as in e.g. Abrusán 2025). From the fact that preposing affects information structure, I infer that preposing in Dutch is not a category-driven operation so it is not restricted by *Move to same category* in (60). In other words, it is not a problem independently that a CP moves to Spec,vP.

- (68) [_{ForceP} Force [_{CP} C [_{TP} subjects T [_{vP} [_{CP} C T v V] $v_{[\textcircled{\text{SCR}}]}$ V _{CP}]]]]
 $\uparrow$
 ?*In die dagen had Jan [_{CP} dat hij DAT zou kopen]₁ nooit* ___₁
 in the day.PL have.3SG.PST Jan that he that should.3SG buy.INF never
gedacht].
 think.PTCP
 ‘In those days, Jan did not think that he should buy THAT.’

In both German and Dutch, the movement of *wh*-phrases and relatives to the Spec,CP position of verb-final embedded clauses (CPs) and the movement of topics and *wh*-phrases to the Spec,ForceP position of V2 clauses (ForcePs) can proceed clause-internally and from infinitival clauses (which was partially discussed in Section 4.3; but see also e.g. Boef 2013; Keine 2016, 2019). Therefore, I assume in both German and Dutch, C can have either [$\textcircled{\text{REL}}$] or [$\textcircled{\text{WH}}$] and Force can have either [$\textcircled{\text{TOP}}$] or [$\textcircled{\text{WH}}$]. It is for space reasons that I do not discuss in detail for all of these constructions in all the Western Germanic dialects whether these movement types can proceed from extraposed/non-extraposed CPs.

Similar to German CPs discussed in Sections 4.2 and 4.3, Dutch CPs are transparent when extraposed. The sentence in (69) exemplifies that *wh*-movement to Spec,ForceP from an extraposed CP is grammatical in Dutch. The difference compared to the German long-topicalization structure in (46) in Section 4.2 is that the Force head in (69) does not have [$\textcircled{\text{TOP}}$] to move topics but [$\textcircled{\text{WH}}$] to move *wh*-phrases. Assuming that Dutch Cs are similar to German Cs in that they do not have [$\textcircled{\text{EFO}}$] but a [$\textcircled{\text{C}}$], (69) can also be analyzed

in the same way as topicalization from extraposed German CPs. After the CP extraposes to the matrix CP, [$\textcircled{\text{WH}}\textcircled{\text{O}}$] on Force can probe along the well-formed sequence constituted by the matrix Force and the embedded C-T- ν -V heads.

- (69) $\left[\text{ForceP } \checkmark \text{ Force}_{[\textcircled{\text{WH}}\textcircled{\text{O}}]} \left[\text{CP } \text{CP} \dots \left[\text{CP } \text{C T } \nu \text{ V } \text{wh} \right] \right] \right]$
 $\left[\text{ForceP } \text{Wat}_2 \text{ had } \text{Jan } \text{nooit } \text{---}_1 \text{ gedacht} \right]$
 what have.3SG.PST Jan never think.PTCP
 $\left[\text{CP } \text{dat } \text{hij } \text{---}_2 \text{ zou } \text{kop-en} \right]_1?$
 that he should.3SG buy-INF
 ‘What has Jan never thought that he should buy?’

While Dutch CPs are transparent to *wh*-movement to Spec,ForceP when extraposed to CP, (70) indicates that preposed CPs are opaque to the same operation (facts originally described by Van Urk and Richards 2015). Given that preposing lands in Spec, ν P, the ungrammaticality of (70) is derived by PAFF: While [$\textcircled{\text{SCR}}\textcircled{\text{O}}$] can find the embedded CP along the well-formed ν -V sequence (i.e. clausal movement can happen), [$\textcircled{\text{WH}}\textcircled{\text{O}}$] on Force fails to find the *wh*-phrase in the preposed clause because it would need to probe along the ill-formed Force-C-T-C-T- ν -V sequence. The latter is not a well-formed *fseq*-fragment because T is not higher than C in *fseq*.

- (70) $\left[\text{ForceP } \text{Force}_{[\textcircled{\text{WH}}\textcircled{\text{O}}]} \text{C T } \left[\nu \text{P } \left[\text{CP } \text{C T } \nu \text{ V } \text{wh} \right] \nu_{[\textcircled{\text{SCR}}\textcircled{\text{O}}]} \text{V CP} \right] \right]$
 $\left[\text{ForceP } \text{Wat}_2 \text{ had } \text{Jan} \right]$
 what have.3SG.PST Jan
 $\left[\text{CP } \text{dat } \text{hij } \text{---}_2 \text{ zou } \text{kop-en} \right]_1 \text{nooit } \text{---}_1 \text{ gedacht}]?$
 that he should.3SG buy-INF never think.PTCP
 Intended: ‘What has Jan never thought that he should buy?’

In sum, the movement of clauses to non-equal-sized clausal projections is possible if the movement operation is not driven by the category of the clause; for example, if the movement is scrambling affecting information structure. Nevertheless, such movement operations to non-equal-sized projections never facilitate sub-extraction to higher projections. The reason why movement to an equal-sized matrix projection fed sub-extraction in

the cases discussed in Sections 4.1, 4.2 and 4.3 was that the sequence of embedded heads found a continuation in the matrix clause: The lowest matrix head c-commanding the highest head of the extraposed embedded clause was the next lowest head in *fseq*. For example, in (69), the matrix Force is the lowest matrix head c-commanding the embedded C. However, with movement to a non-equal-sized matrix projection, the same configuration does not arise because the lowest matrix head c-commanding the extraposition site of an embedded clause is never the next lowest head in the *fseq*. For example, in (70), the matrix T is the lowest matrix head c-commanding the embedded C.¹⁸

4.6 Interim summary and further examples

In this section, I argued that even when β s in a language have no $[\circ EF \circ]$, embedded β Ps can be still escaped if β s in the language lexically have $[\circ \beta \circ]$ features for clausal movement. With such β heads in the lexicon, embedded β Ps are transparent after moving to a matrix β P. That probes c-commanded by the landing site of such embedded β P cannot probe into embedded β P gives us the Williams Cycle pattern.

The introduction discussed that English CPs exhibit Williams Cycle effects (e.g. are opaque to movement to Spec,TP). Here I briefly show that the Williams Cycle-style sub-extraction pattern of English CPs may be analyzed in the fashion of the PAFF-based derivations of this section. For this, one needs to depart from more conventional analyses and propose that no successive cyclic movement is possible from English CPs, that is, English Cs have no $[\circ EF \circ]$. Instead, sub-extraction from English CPs can happen in the same way as from German and Dutch CPs if the English C has $[\circ C \circ]$, which moves the embedded CP making transparent to subsequent sub-extraction. For example, (71) shows that CPs can be extraposed above TP-adjuncts like *yesterday* and such clauses are transparent to *wh*-movement to the matrix Spec,CP (Van Urk and Richards 2015). The PAFF-based analysis of (71) requires $[\circ C \circ]$ to be ordered *before* $[\circ WH \circ]$ on the English C:

Once the clause is extraposed, [_{WH}] probes into the embedded CP along a well formed sequence of embedded heads (C-T-v-V; i.e. in the same way as we saw in the German structure in (55)). Raising to the matrix Spec,TP and Spec,YP (i.e. raising to subject and object, respectively) is impossible from English CPs because CPs become transparent upon moving to the matrix CP.

- (71) [Which team]₁ did you say ₂ yesterday [_{CP} that Arsenal would defeat in the upcoming final ₁]₂?
-

If we choose to analyze English CPs like this, we need to explain why *wh*-movement can proceed from English CPs that linearly occur before temporal adjuncts like *yesterday* and *during the long meeting you had with him last week* in clauses like (72) and (73).

- (72) ? [Which team]₁ did you say [_{CP} that Arsenal would defeat in the upcoming final ₁]
yesterday?
-

- (73) [Which team]₁ did you say [_{CP} that Arsenal would defeat in the upcoming final ₁]
[during the long meeting you had with him last week]?
-

That *wh*-movement in (72) and (73) is acceptable is compatible with the PAFF-based theory if we assume that two syntactic movement operations land in the right-periphery of these sentences, as depicted in (74): After CP is extraposed to the matrix CP via [_{CC}] on C, the adjuncts *yesterday* and *during the long meeting you had with him last week* move to a position above the extraposed clause.

- (74) ...[V _{CP}] [temporal adjunct] _{TP}] CP _{CP}] ✓
-

From the fact that (72) is degraded for my consultants compared to (73) suggests that the movement of these adjuncts is best analyzed as some kind of "heavy constituent shift", which moves sufficiently heavy adjuncts to a CP-adjoined position (a.o. Ross 1967). The marginality of (72) comes from the fact that *yesterday* is not sufficiently heavy phonologically to undergo this rightward movement. This explanation is supported by the fact that my consultants found (73) fully acceptable if *yesterday* was focused or pronounced after

a prosodic break in an appositive manner. That focusing in English makes constituents phonologically heavy enough to shift is discussed among others by Williams (2003).¹⁹

In sum, the English data above is compatible with an analysis where the English C has no $[\circ_{EF}\circ]$ but $[\circ_C\circ]$, that is, similar to Dutch and German CPs, English CPs need to be moved to a matrix CP projection to be transparent. I leave to future research to explore whether there are any further arguments against the purely extraposition-facilitated subextraction from English CPs. If this analysis of the English CPs is on the right track, it holds that English Cs are not like English Ts in that they lack $[\circ_{EF}\circ]$. This also shows that whether successive cyclicity or clausal movement is used should be specified for individual heads rather than entire languages.

Further languages in which the transparency-upon-extraposition analysis of cross-clausal movement may be feasible are Hindi and Hungarian. In these languages, finite clauses must be extraposed from their original, VP-internal position (Mahajan 1990; É. Kiss 2002; Halm 2021; Egressy 2023). Correspondingly, only those operations are possible from the finite clauses of these languages that obey the Williams Cycle, as Keine (2016, 2019, 2020) claims for Hindi and Egressy (2023, 2025, 2026) claims for Hungarian. Due to space limitations, I do not discuss these languages in detail.

There are some attested cases of Williams Cycle effects in Case and Agree (Keine 2016, 2019, 2020; Poole 2022; Egressy 2026). While I do not have space to discuss these in detail, I assume that Agree probes are constrained by PAFF in the same way as movement probes. For example, if a $[\star\phi\star]$ probe occurs on matrix T in a given language and Cs (and Ts) do not have $[\circ_{EF}\circ]$ in the language, $[\star\phi\star]$ cannot reach DPs in the embedded Spec,TP or Spec,vP because it needs to probe along both matrix and embedded heads and $[\circ_C\circ]$ on the matrix C moves the embedded CP too late. In contrast, if agree probes on Force or on C are ordered after $[\circ_C\circ]$ in a given language, they can probe into extraposed CPs, just like the German $[\circ_{TOP}\circ]$ and $[\circ_{WH}\circ]$ probes could do so in Section 4.2. Again, Case can

be handled as Agree (Chomsky 2000, 2001; Legate 2008; Poole 2024).

5 The third option: Neither successive cyclicity nor clausal movement

In Section 3, I discussed the first option for PAFF-compliant cross-clausal movement: If a given category β in a language is associated with $[\circ\text{EF}\circ]$, β Ps can be escaped successive cyclically. In Section 4, I discussed the second option: Although the β s of a language cannot have $[\circ\text{EF}\circ]$, the category β is associated with $[\circ\beta\circ]$ so embedded β Ps can move to equal-sized matrix β Ps whereby they become transparent to Williams Cycle-obeying sub-extraction. The third logical possibility is that a given category β is associated with neither $[\circ\text{EF}\circ]$ nor $[\circ\beta\circ]$ in the lexicon of a language. This results in a scenario when no sub-extraction can proceed from β Ps headed by such β s.

One possible candidate for this third option is Tsez. Polinsky and Potsdam (2001) claim that Tsez has clause-internal scrambling, topicalization, and *wh*-movement to different heights, but these cannot proceed cross-clausally. (75) represents a possible analysis for Tsez embedded clauses, represented here as β Ps under the *fseq* $\alpha > \beta > \gamma > \nu > V$. The matrix probes $[\circ\text{WH}\circ]$, $[\circ\text{TOP}\circ]$, $[\circ\text{SCR}\circ]$ cannot probe into the embedded β P as they have to cross matrix V and then embedded β in a single step, which is not PAFF-compliant because V is the lowest head in the *fseq*. Therefore, the embedded β P cannot be escaped if the embedded β has no $[\circ\text{EF}\circ]$ and the matrix β has no $[\circ\beta\circ]$.²⁰ In other words, Tsez clauses are opaque to sub-extraction for the same reason as unextraposed CPs in German.

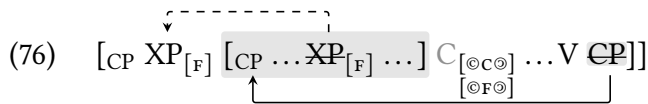
$$(75) \quad [\underbrace{\alpha_{[\circ\text{WH}\circ]} \beta_{[\circ\text{TOP}\circ]} \beta_{[\circ\text{SCR}\circ]} \nu V}_{\text{matrix } \beta\text{P}} [\beta\text{P } \beta \gamma \nu V \text{XP}]]$$

6 Phase unlocking fails to derive Williams Cycle effects

In Section 4, I claimed that under the PAFF-based system, if a given β P is transparent to certain movement operations but opaque to others (i.e. it exhibits selective opacity for sub-extraction), its opacity always follows a Williams Cycle-style pattern: Movement to

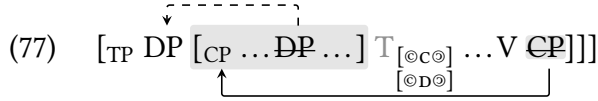
Spec, γ P such that $\beta > \gamma$ in *fseq* is ruled out whereas movement to Spec, α P such that $\alpha > \beta$ in *fseq* is not permitted; whether movement to Spec, β P is possible depends on feature ordering on β . That the Williams Cycle style pattern of sub-extraction is widely attested in the languages of the world was mentioned in the introduction. In this section, I show why a popular extension of Phase Theory, which has been referred to as *phase unlocking* (Rackowski and Richards 2005; Den Dikken 2009; Halpert 2012; Van Urk and Richards 2015), fails to derive that β Ps exhibiting selective opacity for sub-extraction always pattern in the fashion of the Williams Cycle.

The diagram in (76) illustrates a simplified version of phase unlocking which involves clausal movement, as proposed by Van Urk and Richards (2015) (as opposed to exclusively Agree-based unlocking in e.g. Rackowski and Richards 2005). Assuming that the embedded CP in (76) is a phase, no external probe can reach its internal structure. However, once the embedded CP *interacts* with the matrix C head, sub-extraction to the matrix Spec,CP becomes possible. In (76), this interaction emerges because the matrix C has a [$\textcircled{C}\textcircled{C}$] probe, which agrees with the embedded CP and moves it to matrix Spec,CP. After the embedded CP moves to matrix Spec,CP, which can be realized as leftward or rightward movement (i.e. preposing or extraposing, respectively), probes on the matrix C can ignore the phasal boundary constituted by the embedded CP, that is, the CP phase is *unlocked* for probes on C. For example, if the matrix C has an [$\textcircled{F}\textcircled{F}$] probe, XP bearing [F] can move to Spec,CP.²¹

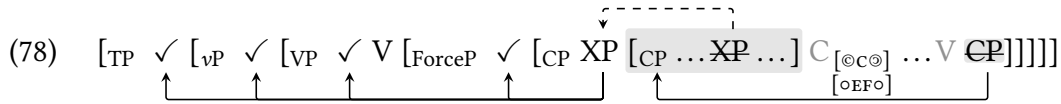


The first reason why phase unlocking cannot derive Williams Cycle effects is that phase unlocking does not constrain in which projection unlocking can happen. For example, the unlocking of a CP phase happens in a TP projection in (77): Under the logic introduced in (76), once the embedded CP is moved by [$\textcircled{C}\textcircled{C}$] on the matrix T, [$\textcircled{D}\textcircled{D}$] on the same T can access DP in CP. That is, phase unlocking by default has no way to derive the ban

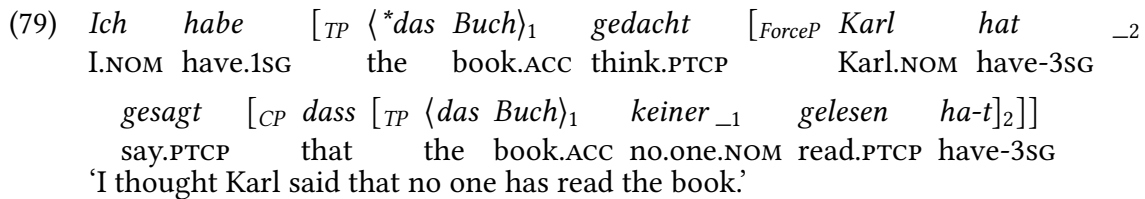
on hyperraising from CP to Spec,TP in (77), whereas hyperraising violates the Williams Cycle. This particular problem is solved by the *Move to same category* theorem introduced in Section 4.4, which allows Cs to have $[\ominus C \ominus]$, as in (76) but not Ts to have $[\ominus C \ominus]$, as in (77).



However, there is a more fundamental problem, as depicted by the three-clausal structure in (78). Once the deepest CP moves to the intermediate CP by $[\ominus C \ominus]$ and XP is sub-extracted to the intermediate Spec,CP by a feature that is denoted here as $[\ominus EF \ominus]$, XP is free to move from this Spec,CP *to any higher projection*, unless we assume that criterial freezing is induced by the probe sub-extracting XP from the deepest CP phase to the intermediate Spec,CP. Thus, XP in (78) can ultimately move from the intermediate Spec,CP not only to intermediate intermediate Spec,ForceP but also to the matrix Spec,VP, Spec,vP or Spec,TP, which violate the Williams Cycle. (For simplicity, (78) assumes that only CPs are phasal.²²)

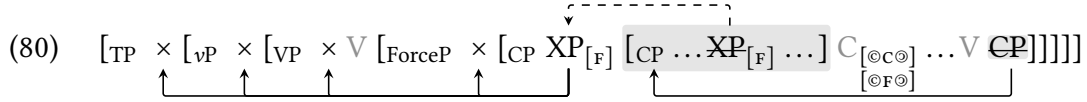


That this is not the correct prediction for German is exemplified by (79): If the deepest CP gets extraposed to the CP projection of an intermediate ForceP clause, scrambling to the matrix Spec,TP is not possible (it does not make scrambling better if use V1 order in the intermediate ForceP).



Thus, without criterial freeizing, phase unlocking cannot derive a pattern in which a given β P is opaque to movement to any Spec, γ P such that $\beta > \gamma$ in *fseq*. On the other hand, assuming that all movement probes on a given β induce criterial freezing makes

the opacity of β P's more restrictive than the Williams Cycle pattern. In (80), the deepest CP is unlocked in an intermediate CP after moving to Spec,CP, which is followed by the sub-extraction of XP to the specifier of the intermediate CP. Assuming that sub-extraction is done via $[\odot_F \odot]$ inducing criterial freezing, the further movement of XP is not only ruled out to the Spec,TP/Spec,vP positions of higher clauses but also to the clause-internal Spec,ForceP. This is clearly not the sub-extraction pattern we saw in Sections 4.2 and 4.3 for German and in Section 4.5 for Dutch; in those languages, an embedded CP extraposed to CP is transparent to movement to Spec,ForceP.



In sum, as shown in (80), phase unlocking can derive the selective transparency of a CP phase to movement to Spec,CP, if we assume the possibility of criterial freezing and require that clauses move to equal-sized matrix projections. However, phase unlocking has a hard time deriving a pattern where movement to both Spec,CP and Spec,ForceP is possible but movement to Spec,TPs and Spec,vPs is not, as shown in (78). The general problem is that in Phase Theory, once sub-extraction happens from a phase in a way that the sub-extracted XP is not frozen, we lose track of where XP originates. One theory trying to deal with this "keeping track" problem is the *buffer*-based theory by Müller (2014a,b) mentioned in Section 2.2.

7 Conclusion

My proposal was composed from three crucial ingredients while unifying successive cyclicity, Williams Cycle effects, and Williams Cycle obviation. Firstly, I proposed criterial and edge features are lexically specified for individual heads in individual languages (a.k.a. the Borer-Chomsky Conjecture in Borer 1984 and Chomsky 1995). For example, I argued that the P'urhepecha C and the English T have $[\odot_{EF} \odot]$ whereas German, English, and

Dutch C heads have no $[\circ EF \circ]$ but $[\circ C \circ]$. Secondly, I proposed that the category-driven movement of embedded clauses always happens to equal-sized matrix projections. For example, the category-based movement of CPs can land in the matrix CP but not in the matrix TP or ForceP. Finally, I proposed that probing is constrained by a single locality constraint, PAFF, which dictates probing must proceed along well-formed *fseq*-fragments.

The PAFF-based theory predicts that in any given language, each category β exhibits one of the options in (81). I assumed that if a given category β in a language is associated with $[\circ EF \circ]/[\circ \beta \circ]$, all β heads have the same $[\circ EF \circ]/[\circ \beta \circ]$ features in that language. If a given category β exhibits Option 1 and has an $[\circ EF \circ]$ feature, movement can proceed from β P's successive cyclically and Williams Cycle effects are obviated for sub-extraction from such β P's. Section 3 used the English TP and P'urhepecha CP to illustrate this pattern. Under Option 2, heads of a given β category have no $[\circ EF \circ]$ but have $[\circ \beta \circ]$ so an embedded β P can become transparent upon merging with a matrix β P; under Option 2, sub-extraction obeys the Williams Cycle. Section 4 primarily used German CP and Dutch CPs to illustrate this behavior, but English CPs can also be argued to exhibit this pattern. Finally, there is Option 3: it is also possible that β s are associated with neither $[\circ EF \circ]$ nor $[\circ \beta \circ]$; in this case, no movement can proceed from β P. As discussed in Section 5, embedded clauses in Tsez are a candidate for this option.

(81)		$[\circ EF \circ]$	$[\circ \beta \circ]$	Prediction for sub-extraction pattern
	Option 1	✓	(irrelevant)	No opacity
	Option 2	×	✓	Williams Cycle obeying extraction only
	Option 3	×	×	Complete opacity

The PAFF-based system makes very clear empirical predictions. In particular, the absence of Williams Cycle effects is predicted to always correspond to no opacity: If a constituent can be escaped successive cyclically, all movement types can proceed from it. On the other hand, selective opacity is always Williams Cycle-like: Probes lower than β cannot, whereas probes higher β can look into a β P that is made transparent via movement

to an equal-sized matrix projection.

The PAFF-based theory also predicts that successive cyclic movement and Williams Cycle effects are in a complementary distribution. Therefore, projections for which the two have been simultaneously claimed need to be reanalyzed. The only projection that I am aware that has been argued to exhibit Williams Cycle effects *and* reflexes of successive cyclicity is the German ForceP (as analyzed by Keine 2016, 2020). For German ForcePs, the reflex of successive cyclicity is the V1 order in ForcePs from which long movement to Spec,ForceP has taken place; ForcePs without sub-extraction are V2. One way to re-analyze German ForcePs is to say that V1 clauses are Force1Ps, whereas V2 clauses are Force2Ps, where $\text{Force2} > \text{Force1}$ in *fseq*, and both Force1 and Force2 lack $[\circ \text{EF} \circ]$. That is, topicalization can only proceed from smaller clauses, Force1Ps, but not from larger clauses, Force2Ps. Whether this re-analysis is feasible is left for further research.

In addition to providing a unified analysis for successive cyclicity, Williams Cycle effects, and Williams Cycle obviation, PAFF makes a number of concepts and assumptions unnecessary in the syntactic framework. Most importantly, under the PAFF-based theory, phases and locality domains in general do not need to be established as independent concepts. In this respect, the PAFF-based framework joins the line of other frameworks with the purpose of dispensing phases from syntactic theory (Halpert and Zeijlstra 2024; Keine and Zeijlstra 2025). Still, the PAFF-based theory is compatible with clause-internal stops in (for example) Spec,vP, as argued in the Appendix.

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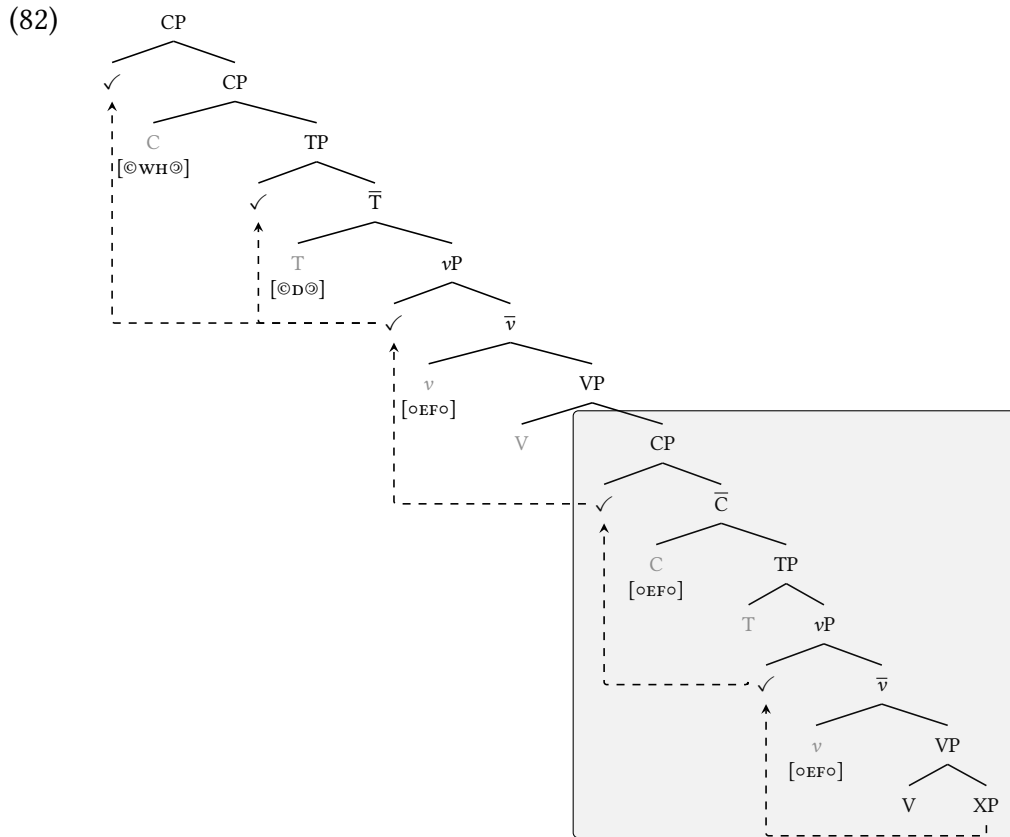
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Appendix: Successive cyclicity via Spec, ν Ps

Among other languages, Dinka exhibits reflexes of clause-internal successive cyclic steps in Spec, ν Ps as well as reflexes of successive cyclic steps in Spec, CPs (i.e. clause edges) (Van Urk and Richards 2015; Urk 2015). Previous work on selective opacity could not handle selective opacity and intermediate stops in Spec, ν Ps in the same framework (a.o. Keine 2016, 2019, 2020; Poole 2022). In contrast, my analysis is compatible with intermediate movement steps in Spec, ν P. For example, assume that both C and ν (but no other heads) have $[\circ_{EF}\circ]$ in a language, as depicted in (82). Within the embedded clause, $[\circ_{EF}\circ]$ on ν can reach an XP in the V-complement position along the well-formed ν -V sequence, and the $[\circ_{EF}\circ]$ on C can reach XP in Spec, ν P along the well-formed C-T sequence. $[\circ_{EF}\circ]$ on the matrix ν can access XP in the embedded Spec, CP along the well-formed ν -V sequence.

Alternatively, if v has $[\textcircled{D}\textcircled{\circ}]$, it can also reach XP along the v -V sequence. In general, XP can move to any position from the embedded Spec,CP or the matrix Spec,vP (i.e. there is no Williams Cycle-style constraint on movement). For example, $[\textcircled{D}\textcircled{\circ}]$ on the matrix T can reach XP, it only needs to cross the T head and $[\textcircled{\text{WH}}\textcircled{\circ}]$ on the matrix C can reach XP along the C-T sequence.



Notes

¹For reflexes of successive cyclicity, see e.g. McCloskey 1979; Chung 1982; Torrence 2013, Georgi 2014; for various phase-based accounts of successive cyclicity, see e.g. Chomsky 2000; 2001; Heck and Müller 2000; Fox and Pesetsky 2005; Bošković 2007; Van Urk and Richards 2015.

²The domination-based formulation of (16): Within the domain dominated by the maximal projection of the head that the probe is located on, consider the projections which dominate the goal with all their

segments. (The probe's projection is considered iff it dominates the goal with every segment.) The categories of the considered projections must make up a well-formed *fseq*-fragment.

³PAFF is a more general version of *Move Along Increasing Fseq-indices* (MAIF) proposed by Egressy (2023). The difference between the two is that while MAIF constraints movement only, PAFF constrains any kind of probing.

⁴The absence of superiority effects for $[\circ_{EF}\circ]$ may follow from the property that $[\circ_{EF}\circ]$ does not search for goals with specific features. Rizzi's *Featural Relativized Minimality* (Rizzi 1990; Starke 2001; Rizzi 2004, 2018:p.347) says that in an $X...Z...Y$ configuration, where X is a probe that c-commands Z and Y and Z asymmetrically c-commands Y , Z is an intervener between Z and X iff " Z matches X in terms of Relevant Syntactic Features." Since $[\circ_{EF}\circ]$ is not specified for any feature, no Z can act as an intervener in $[\circ_{EF}\circ]$'s search because no Z can have a feature that matches the features of $[\circ_{EF}\circ]$.

⁵Further arguments for genuine sub-extraction have been brought by Postal (1974) and Lasnik and Saito (1999). At the same time, Neeleman and Payne (2020) have recently questioned the arguments for cross-clausal raising.

⁶According to Zyman (2018), subject positions in P'urhepecha can vary but it is always a Spec, β P such that $C > \beta \geq v$ in *fseq*.

⁷For another instance of the "single edge phenomenon", see Aravind (2021).

⁸The argumentation of this section is partially based on the ideas by Egressy (2023).

⁹While Müller claims that structures like (39) are grammatical, not all German speakers accept in-situ CPs.

¹⁰Uncited German data in Sections 4.2, 4.3, and 6 comes from personal communication with Stefan Keine, Martin Salzmann, Jutta Hartmann, Anke Himmelreich, and Katharina Hartmann. Uncited Dutch in Section 4.5 data comes from personal communication with Melle Groen, Hedde Zeijlstra, and Sjeff Barbiers. In order to test the grammaticality of English sentences in Section 4.6, I asked 11 native speakers of North American English.

¹¹It needs to be highlighted that the movement of embedded clauses is criterial, that is, C has $[\circ_C\circ]$, not $[\bullet_C\bullet]$, as shown by (83). After a CP extraposes and *wh*-movement has proceeded from it, as in sentences like (52), the CP cannot be topicalized after extraposition and *wh*-subextraction, even though German CPs can be topicalized otherwise (Müller 1999:p.34).

- (83) $*[_{CP} \text{ Dass Fritz } _1 \text{ liebt}]_2 \text{ weiß ich nicht } [_{CP} \text{ wen}_1 \text{ er } _2 \text{ gesagt hat } _2]$.
 that Fritz love.3SG know.1SG I not who.ACC he said have.3SG
 Intended: 'I do not know who he said that Fritz loves.'

Furthermore, German has a construction that involves the topicalization of VPs/vPs with a V-CP order (see e.g. Haider 2010). While this requires the CP to move into a low, VP/vP-internal position, I assume that this clausal movement is some kind of scrambling rather than genuine category-driven movement. I am thankful to Anke Himmelreich for bringing this construction to my attention.

¹²For my analysis, it would not make too much difference to assign [$\textcircled{\text{SCR}}$] to v in German (as in Müller 2014a,b).

¹³German exhibits a construction usually referred to as "partial *wh*-movement" or "*wh*-scope marking": the *wh*-phrase originating in the embedded clause does not move to the matrix Spec,CP only to the embedded Spec,CP, while its matrix scope is marked by *was* 'what' in the matrix Spec,CP (Mcdaniel 1989). While Mcdaniel (1989) proposes that this is evidence for successive cyclic movement in which the intermediate copy in the embedded Spec,CP is pronounced, I follow here the *indirect dependency analysis* of *wh*-scope-marking by Dayal (1994) on Hindi, Herburger (1994) on German, and Horvath (1997) on Hungarian. These analyses propose that the *wh*-phrase in the embedded Spec,CP does not form a chain with the *wh*-phrase in the matrix Spec,CP. Instead, the *wh*-phrase in the matrix Spec,CP is associated with the entire embedded clause. That is, we can assume that *wh*-movement both the embedded and the matrix Spec,CP is driven by [$\textcircled{\text{WH}}$] and no [$\textcircled{\text{EF}}$] is involved.

¹⁴While the standard analysis of sentential subjects involves a CP (re-)merging with a CP in Spec,TP, Hartman (2012) argues that sentential subjects always occur within DPs, whereby T does not merge with CP but with a DP.

¹⁵I am thankful to Hedde Zeijlstra (p.c.) for suggesting this angle on the problem.

¹⁶Whether a head whose *fseq* index is n can move a category whose *fseq* index is $n-1$ (i.e. the category one-step lower) is less certain. While such movement (e.g. C moving TP from its complement to its specifier) necessarily violates anti-locality (Abels 2012), this is exactly the movement step that has been proposed by the antisymmetry-compliant derivation of the head-final order (e.g. Kayne 1994; Sheehan et al. 2017). Additionally, in a theory where internal and external merge are driven by the same kind of features, selection for lower categories cannot be excluded because that would exclude basic clause building. Hence, I do not propose a principle parallel to (61) to exclude the selection/movement of lower categories.

¹⁷The ungrammaticality of *wh*-movement from preposed CPs in Dutch would also work if the scrambling was assumed to land in a lower Spec,TP, where the higher Spec,TP is occupied by subjects.

¹⁸One configuration in which movement to a non-equal-sized matrix projection is predicted to feed PAFF-compliant sub-extraction follows the structure of (63) with the difference that the movement of the clause is not category-based (e.g. CP gets scrambled to Spec,TP, which is followed by hyperraising to a

higher Spec,TP). In such structures, the clause is moved to a matrix clausal projection and sub-extraction lands within the same matrix projection, that is, the probe driving the sub-extraction does not probe from a matrix head but a higher projection so it does not have to consider any matrix heads but can probe along embedded heads exclusively. While I cannot bring any specific examples, pied-piping with inversion may be analyzed in this fashion (for examples, see Heck 2008): A possibly larger embedded constituent undergoes movement to the specifier of a matrix projection of a lower *fseq*-category, and we sub-extract from the larger constituent to a higher matrix specifier within the same projection.

¹⁹The rightward movement of temporal adjuncts may be in an option in German and Dutch, too; temporal adjuncts are fine in a position after the extraposed CP. This can be achieved by the CP-adjunction of temporal adjuncts, just as it is demonstrated in the main text for English. This does not change the argument in Sections 4.2 and 4.5; the central point is that while temporal adjuncts *can* show up in the TP-adjoined position, preceding extraposed CPs, *because*-clauses cannot.

²⁰Using the heads used by Polinsky and Potsdam (2001), [$\textcircled{\text{WH}}$] is on a C head and [$\textcircled{\text{TOP}}$] is on a Top head, and [$\textcircled{\text{SCR}}$] is on Top or a lower head; the embedded clauses of Tsez are TopP or CP-sized, but in any case opaque to movement. Polinsky and Potsdam add that absolutive arguments topicalized to the highest specifier of TopP-sized embedded clauses (here depicted as Spec, β P) can trigger noun-class agreement on the matrix verb (Polinsky and Potsdam 2001:p.605-606). The PAFF-based theory predicts this: the [$\star\phi\star$] probe on matrix *v* can probe along the *v*-V sequence to reach the highest specifier of a β P, [$\star\phi\star$] does not have to cross any embedded heads. Since topicalization to Spec, β P induces criterial freezing, it can feed Agree probes but not movement probes. Essentially, once the topicalized absolutive occurs deeper, e.g. an intermediate TopP clause or a clause-internal CP layer is present above it (e.g. Polinsky and Potsdam 2001:p.618), Long-Distance Agreement fails in Tsez because the [$\star\phi\star$] probe does not only cross matrix heads but also the heads of the intermediate clause. While Polinsky and Potsdam (2001) mention clauses with the same opacity showing up to the right of the matrix verb, these are very uncommon and the exact motivation of the movement is unknown according to Maria Polinsky (p.c.). Here I assume that even if these clauses are moved to the right, this movement is operator movement rather than category-driven movement, which does not happen to an equal-sized projection, that is, the movement of Tsez clauses is similar to the leftward movement of Dutch CPs discussed in Section 4.5.

²¹Certain versions of the theory require the same probe on the matrix C to interact with the phasal CP and the XP to be sub-extracted (e.g. Rackowski and Richards 2005), that is, it is not sufficient if the probes occur on the same head. Here I assume that a phase is unlocked for all the probes that occur on the same

head as the probe that interacted with the phase (as in Van Urk and Richards 2015).

²²Irrespectively of the phasality of νP , from the intermediate Spec,CP in (78), XP can move to matrix Spec,VP or some Spec, γP such that $\nu > \gamma > V$ in *fseq*, possibly taking a stop in Spec,ForceP; to my knowledge, Force and V are typically not assumed to be phases. Movement to Spec,VP or some Spec,Spec, γP such that $\nu > \gamma > V$ is still a violation of the Williams Cycle.